

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 13/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTMwODUxNTRfMjc3>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

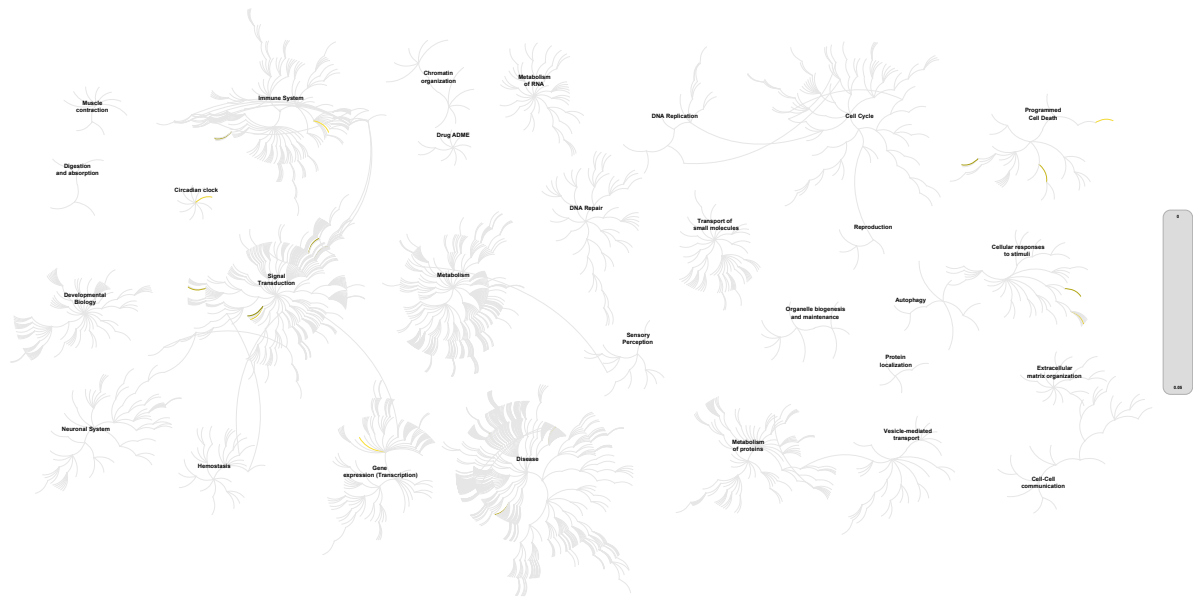
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 49 out of 56 identifiers in the sample were found in Reactome, where 1228 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTMwODUxNTRfMjc3. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Nuclear Receptor transcription pathway	9 / 126	0.005	4.75e-08	6.30e-05	2 / 2	1.26e-04
Interleukin-4 and Interleukin-13 signaling	14 / 473	0.019	3.41e-04	0.226	9 / 47	0.003
CASP8 activity is inhibited	2 / 13	5.24e-04	0.002	0.695	2 / 2	1.26e-04
The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex	2 / 21	8.46e-04	0.006	0.695	3 / 4	2.52e-04
NFE2L2 regulating anti-oxidant/detoxification enzymes	7 / 145	0.006	0.009	0.695	5 / 28	0.002
Erythropoietin activates Phosphoinositide-3-kinase (PI3K)	2 / 29	0.001	0.012	0.695	7 / 7	4.41e-04
Defective SLC11A2 causes hypochromic microcytic anemia, with iron overload 1 (AHMIO1)	1 / 3	1.21e-04	0.017	0.695	1 / 1	6.30e-05
Caspase activation via Death Receptors in the presence of ligand	3 / 42	0.002	0.023	0.695	7 / 9	5.67e-04
Adenosine P1 receptors	1 / 5	2.01e-04	0.027	0.695	1 / 2	1.26e-04
UNC93B1 deficiency - HSE	1 / 6	2.42e-04	0.033	0.695	1 / 1	6.30e-05
Regulation of HMOX1 expression and activity	2 / 51	0.002	0.033	0.695	4 / 4	2.52e-04
IRS activation	1 / 7	2.82e-04	0.038	0.695	3 / 3	1.89e-04
The NLRP1 inflammasome	1 / 7	2.82e-04	0.038	0.695	2 / 4	2.52e-04
SMAC(DIABLO)-mediated dissociation of IAP:caspase complexes	1 / 7	2.82e-04	0.038	0.695	1 / 3	1.89e-04
Signaling by MAPK mutants	1 / 8	3.22e-04	0.043	0.695	1 / 1	6.30e-05
PI3K/AKT activation	2 / 63	0.003	0.049	0.695	4 / 5	3.15e-04
Erythropoietin activates STAT5	1 / 9	3.62e-04	0.049	0.695	3 / 3	1.89e-04
SOS-mediated signalling	1 / 10	4.03e-04	0.054	0.695	2 / 2	1.26e-04
Defective Mismatch Repair Associated With MSH2	1 / 10	4.03e-04	0.054	0.695	1 / 2	1.26e-04
BH3-only proteins associate with and inactivate anti-apoptotic BCL-2 members	4 / 70	0.003	0.058	0.695	3 / 5	3.15e-04
RSK activation	1 / 11	4.43e-04	0.059	0.695	4 / 4	2.52e-04
Activated NTRK2 signals through PI3K	1 / 11	4.43e-04	0.059	0.695	2 / 2	1.26e-04

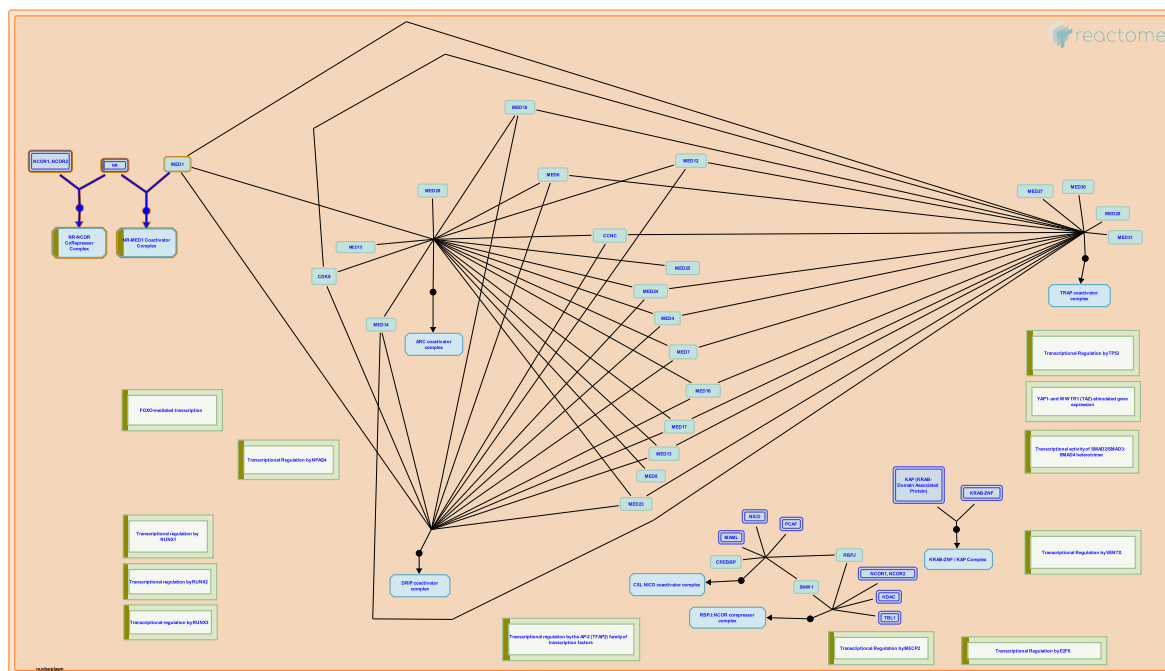
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Activation of NOXA and translocation to mitochondria	1 / 11	4.43e-04	0.059	0.695	3 / 5	3.15e-04
Expression and Processing of Neurotrophins	1 / 11	4.43e-04	0.059	0.695	2 / 5	3.15e-04
NGF processing	1 / 11	4.43e-04	0.059	0.695	2 / 5	3.15e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Nuclear Receptor transcription pathway (R-HSA-383280)



A classic example of bifunctional transcription factors is the family of Nuclear Receptor (NR) proteins. These are DNA-binding transcription factors that bind certain hormones, vitamins, and other small, diffusible signaling molecules. The non-liganded NRs recruit specific corepressor complexes of the NCOR/SMRT type, to mediate transcriptional repression of the target genes to which they are bound. During signaling, ligand binding to a specific domain the NR proteins induces a conformational change that results in the exchange of the associated CoR complex, and its replacement by a specific coactivator complex of the TRAP / DRIP / Mediator type. These coactivator complexes typically nucleate around a MED1 coactivator protein that is directly bound to the NR transcription factor.

A general feature of the 49 human NR proteins is that in the unliganded state, they each bind directly to an NCOR corepressor protein, either NCOR1 or NCOR2 (NCOR2 was previously named "SMRT"). This NCOR protein nucleates the assembly of additional, specific corepressor proteins, depending on the cell and DNA context. The NR-NCOR interaction is mediated by a specific protein interaction domain (PID) present in the NRs that binds to specific cognate PID(s) present in the NCOR proteins. Thus, the human NRs each take part in an NR-NCOR binding reaction in the absence of binding by their ligand.

A second general feature of the NR proteins is that they each contain an additional, but different PID that mediates specific binding interactions with MED1 proteins. In the ligand-bound state, NRs each take part in an NR-MED1 binding reaction to form an NR-MED1 complex. The bound MED1 then functions to nucleate the assembly of additional specific coactivator proteins, depending on the cell and DNA context, such as what specific target gene promoter they are bound to, and in what cell type.

The formation of specific MED1-containing coactivator complexes on specific NR proteins has been well-characterized for a number of the human NR proteins (see Table 1 in (Bourbon, 2004)). For example, binding of thyroid hormone (TH) to the human TH Receptor (THRA or THRB) was found to result in the recruitment of a specific complex of Thyroid Receptor Associated Proteins - the TRAP coactivator complex - of which the TRAP220 subunit was later identified to be the Mediator 1 (MED1) homologue.

Similarly, binding of Vitamin D to the human Vitamin D3 Receptor was found to result in the recruitment of a specific complex of D Receptor Interacting Proteins - the DRIP coactivator complex, of which the DRIP205 subunit was later identified to be human MED1.

References

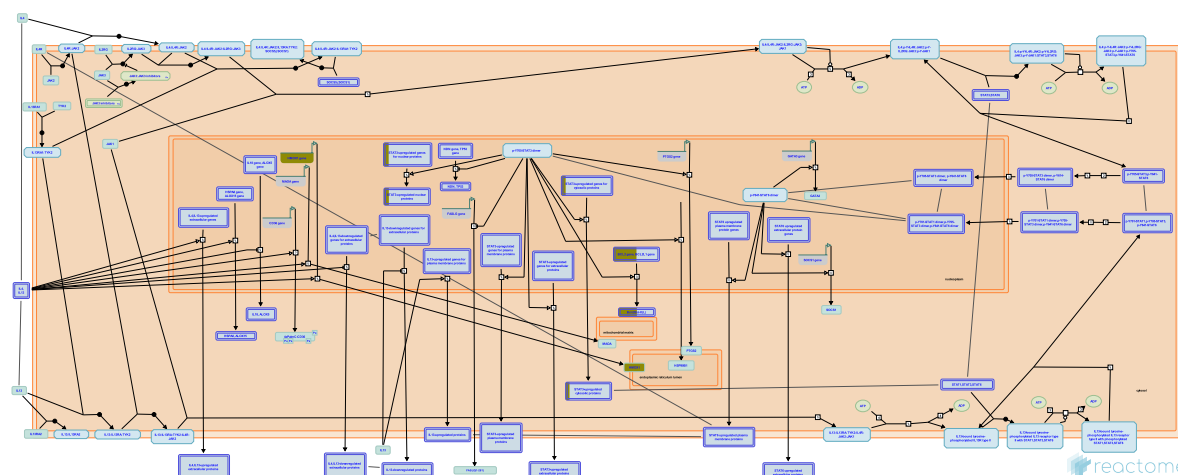
Edit history

Date	Action	Author
2008-11-20	Authored	Caudy M
2008-12-03	Created	Caudy M
2009-05-27	Edited	Caudy M
2009-08-29	Reviewed	Freedman LP

1 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9

2. Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)



Interleukin-4 (IL4) is a principal regulatory cytokine during the immune response, crucially important in allergy and asthma (Nelms et al. 1999). When resting T cells are antigen-activated and expand in response to Interleukin-2 (IL2), they can differentiate as Type 1 (Th1) or Type 2 (Th2) T helper cells. The outcome is influenced by IL4. Th2 cells secrete IL4, which both stimulates Th2 in an autocrine fashion and acts as a potent B cell growth factor to promote humoral immunity (Nelms et al. 1999).

Interleukin-13 (IL13) is an immunoregulatory cytokine secreted predominantly by activated Th2 cells. It is a key mediator in the pathogenesis of allergic inflammation. IL13 shares many functional properties with IL4, stemming from the fact that they share a common receptor subunit. IL13 receptors are expressed on human B cells, basophils, eosinophils, mast cells, endothelial cells, fibroblasts, monocytes, macrophages, respiratory epithelial cells, and smooth muscle cells, but unlike IL4, not T cells. Thus IL13 does not appear to be important in the initial differentiation of CD4 T cells into Th2 cells, rather it is important in the effector phase of allergic inflammation (Hershey et al. 2003).

IL4 and IL13 induce “alternative activation” of macrophages, inducing an anti-inflammatory phenotype by signaling through IL4R alpha in a STAT6 dependent manner. This signaling plays an important role in the Th2 response, mediating anti-parasitic effects and aiding wound healing (Gordon & Martinez 2010, Loke et al. 2002)

There are two types of IL4 receptor complex (Andrews et al. 2006). Type I IL4R (IL4R1) is predominantly expressed on the surface of hematopoietic cells and consists of IL4R and IL2RG, the common gamma chain. Type II IL4R (IL4R2) is predominantly expressed on the surface of nonhematopoietic cells, it consists of IL4R and IL13RA1 and is also the type II receptor for IL13. (Obiri et al. 1995, Aman et al. 1996, Hilton et al. 1996, Miloux et al. 1997, Zhang et al. 1997). The second receptor for IL13 consists of IL4R and Interleukin-13 receptor alpha 2 (IL13RA2), sometimes called Interleukin-13 binding protein (IL13BP). It has a high affinity receptor for IL13 (Kd = 250 pmol/L) but is not sufficient to render cells responsive to IL13, even in the presence of IL4R (Donaldson et al. 1998). It is reported to exist in soluble form (Zhang et al. 1997) and when overexpressed reduces JAK-STAT signaling (Kawakami et al. 2001). It's function may be to prevent IL13 signalling via the functional IL4R:IL13RA1 receptor. IL13RA2 is overexpressed and enhances cell invasion in some human cancers (Joshi & Puri 2012).

The first step in the formation of IL4R1 (IL4:IL4R:IL2RB) is the binding of IL4 with IL4R (Hoffman et al. 1995, Shen et al. 1996, Hage et al. 1999). This is also the first step in formation of IL4R2 (IL4:IL4R:IL13RA1). After the initial binding of IL4 and IL4R, IL2RB binds (LaPorte et al. 2008), to form IL4R1. Alternatively, IL13RA1 binds, forming IL4R2. In contrast, the type II IL13 complex (IL13R2) forms with IL13 first binding to IL13RA1 followed by recruitment of IL4R (Wang et al. 2009).

Crystal structures of the IL4:IL4R:IL2RG, IL4:IL4R:IL13RA1 and IL13:IL4R:IL13RA1 complexes have been determined (LaPorte et al. 2008). Consistent with these structures, in monocytes IL4R is tyrosine phosphorylated in response to both IL4 and IL13 (Roy et al. 2002, Gordon & Martinez 2010) while IL13RA1 phosphorylation is induced only by IL13 (Roy et al. 2002, LaPorte et al. 2008) and IL2RG phosphorylation is induced only by IL4 (Roy et al. 2002).

Both IL4 receptor complexes signal through Jak/STAT cascades. IL4R is constitutively-associated with JAK2 (Roy et al. 2002) and associates with JAK1 following binding of IL4 (Yin et al. 1994) or IL13 (Roy et al. 2002). IL2RG constitutively associates with JAK3 (Boussiotis et al. 1994, Russell et al. 1994). IL13RA1 constitutively associates with TYK2 (Umeshita-Suyama et al. 2000, Roy et al. 2002, LaPorte et al. 2008, Bhattacharjee et al. 2013).

IL4 binding to IL4R1 leads to phosphorylation of JAK1 (but not JAK2) and STAT6 activation (Takeda et al. 1994, Ratthe et al. 2007, Bhattacharjee et al. 2013).

IL13 binding increases activating tyrosine-99 phosphorylation of IL13RA1 but not that of IL2RG. IL4 binding to IL2RG leads to its tyrosine phosphorylation (Roy et al. 2002). IL13 binding to IL4R2 leads to TYK2 and JAK2 (but not JAK1) phosphorylation (Roy & Cathcart 1998, Roy et al. 2002).

Phosphorylated TYK2 binds and phosphorylates STAT6 and possibly STAT1 (Bhattacharjee et al. 2013).

A second mechanism of signal transduction activated by IL4 and IL13 leads to the insulin receptor substrate (IRS) family (Kelly-Welch et al. 2003). IL4R1 associates with insulin receptor substrate 2 and activates the PI3K/Akt and Ras/MEK/Erk pathways involved in cell proliferation, survival and translational control. IL4R2 does not associate with insulin receptor substrate 2 and consequently the PI3K/Akt and Ras/MEK/Erk pathways are not activated (Busch-Dienstfertig & González-Rodríguez 2013).

References

- Nelms K, Keegan AD, Zamorano J, Ryan JJ & Paul WE (1999). The IL-4 receptor: signaling mechanisms and biologic functions. *Annu. Rev. Immunol.*, 17, 701-38. [↗](#)
- Hershey GK (2003). IL-13 receptors and signaling pathways: an evolving web. *J. Allergy Clin. Immunol.*, 111, 677-90; quiz 691. [↗](#)

Edit history

Date	Action	Author
2015-07-01	Authored	Jupe S
2015-07-01	Created	Jupe S
2016-09-02	Edited	Jupe S
2016-09-02	Reviewed	Leibovich SJ

5 submitted entities found in this pathway, mapping to 10 Reactome entities

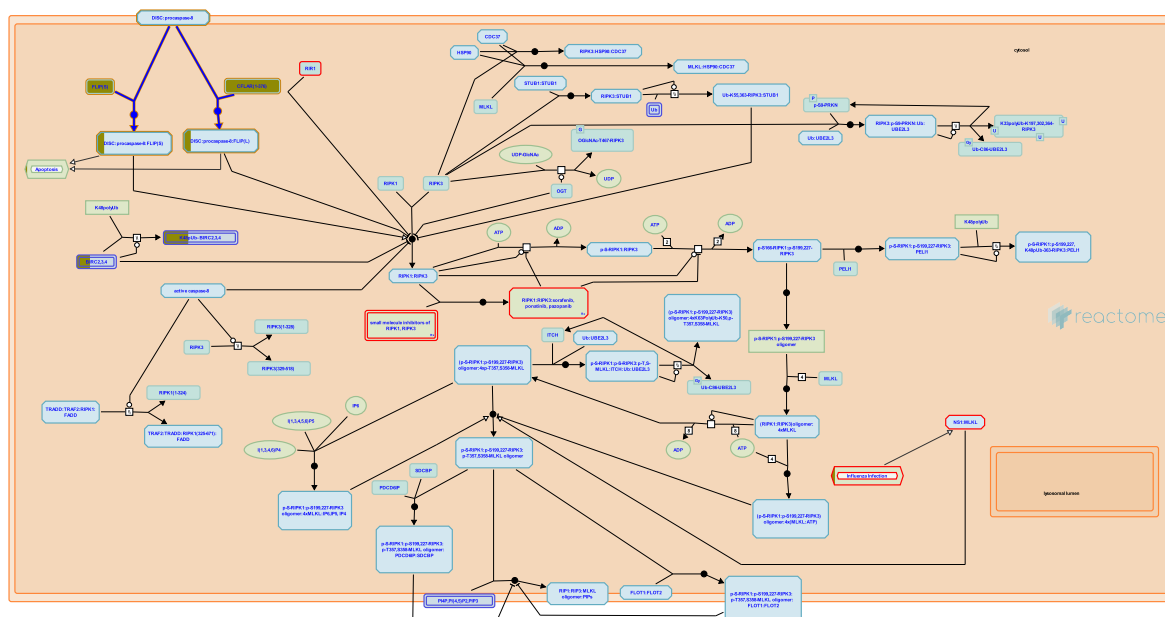
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BCL2L1	Q07817	HMOX1	P09601	MYC	P01106
PIK3R1	P27986	ZEB1	P37275		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
BCL2L1	ENSG00000171552	HMOX1	ENSG00000100292	MYC	ENSG00000136997
PIK3R1	ENSG00000145675	ZEB1	ENSG00000148516		

Interactors found in this pathway (4)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
IFNGR1	P15260	P23458	MYC	EBI-1265559	P23771
PIK3R1	P27986-2, P27986	P48023, O15524	SNX30	Q8WV41	P48023

3. CASP8 activity is inhibited (R-HSA-5218900)



Cell death triggered by extrinsic stimuli via death receptors or toll-like receptors (e.g., TLR3, TLR4) may result in either apoptosis or regulated necrosis (necroptosis) (Holler N et al. 2000; Kalai M et al. 2002; Kaiser WJ and Offermann MK 2005; Yang P et al. 2007). Caspase-8 (CASP8) is a cysteine protease, which functions as a key mediator for determining which form of cell death will occur (Kalai M et al. 2002). The proteolytic activity of a fully processed, heterotetrameric form of CASP8 in human and rodent cells is required for proapoptotic signaling and also for a cleavage of kinases RIPK1 and RIPK3, while at the same time preventing RIPK1/RIPK3-dependent regulated necrosis (Juo P et al. 1998; Lin Y et al. 1999; Holler N et al. 2000; Hopkins-Donaldson S et al. 2000). A blockage of CASP8 activity in the presence of caspase inhibitors such as Z-VAD-FMK (pan-caspase inhibitor), endogenous FLIP(S) or viral FLIP-like protein was found to switch signaling to necrotic cell death (Thome M et al. 1997; Kalai M et al. 2002; Feoktistova M et al. 2011; Sawai H 2013).

References

- Thome M, Schneider P, Hofmann K, Fickenscher H, Meinel E, Neipel F, ... Tschopp J (1997). Viral FLICE-inhibitory proteins (FLIPs) prevent apoptosis induced by death receptors. *Nature*, 386, 517-21. [🔗](#)
- Sawai H (2013). Differential effects of caspase inhibitors on TNF-induced necroptosis. *Biochem. Biophys. Res. Commun.*, 432, 451-5. [🔗](#)
- Kalai M, Van Loo G, Vanden Berghe T, Meeus A, Burm W, Saelens X & Vandenabeele P (2002). Tipping the balance between necrosis and apoptosis in human and murine cells treated with interferon and dsRNA. *Cell Death Differ.*, 9, 981-94. [🔗](#)
- Feoktistova M, Geserick P, Kellert B, Dimitrova DP, Langlais C, Hupe M, ... Leverkus M (2011). cIAPs block Ripoptosome formation, a RIP1/caspase-8 containing intracellular cell death complex differentially regulated by cFLIP isoforms. *Mol. Cell*, 43, 449-63. [🔗](#)

Edit history

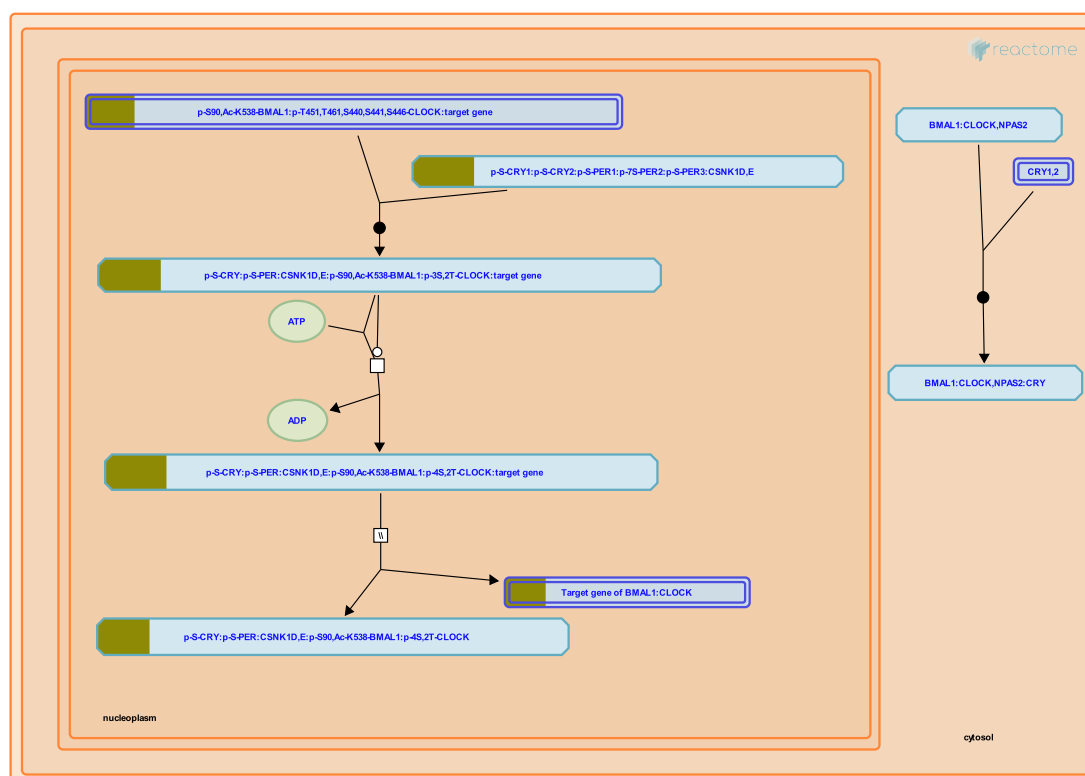
Date	Action	Author
2013-12-20	Authored	Shamovsky V

Date	Action	Author
2013-12-20	Created	Shamovsky V
2014-10-31	Reviewed	Gillespie ME
2015-02-10	Edited	Shamovsky V
2015-02-15	Reviewed	Chan FK
2020-08-28	Reviewed	Murphy JM

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
CFLAR	O15519-1, O15519-2

4. The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex (R-HSA-9931521)



A complex containing Cryptochrome proteins CRY1, CRY2, Period proteins PER1, PER2, PER3, and kinases CSNK1D and/or CSNK1E binds a heterodimer containing phosphorylated CLOCK and phosphorylated, acetylated BMAL1 bound to E-boxes in the promoters of target genes (inferred from mouse homologs in Griffin et al. 1999, Langmesser et al. 2008, Ye et al. 2011, Ye et al. 2014, Aryal et al. 2017, Cao et al. 2021). This effectively delivers the kinases CSNK1D and/or CSNK1E to the proximity of CLOCK, resulting in further phosphorylation of CLOCK (inferred from mouse homologs in Cao et al. 2021). NPAS2, which acts redundantly with CLOCK, may be similarly phosphorylated. The phosphorylation of CLOCK causes the dissociation of the BMAL1:CLOCK heterodimer from DNA (inferred from mouse homologs in Ye et al. 2014, Chiou et al. 2016, Cao et al. 2021), terminating the diurnal activation of gene expression by BMAL1:CLOCK.

References

- Ye R, Selby CP, Ozturk N, Annayev Y & Sancar A (2011). Biochemical analysis of the canonical model for the mammalian circadian clock. *J Biol Chem*, 286, 25891-902. [🔗](#)
- Aryal RP, Kwak PB, Tamayo AG, Gebert M, Chiu PL, Walz T & Weitz CJ (2017). Macromolecular Assemblies of the Mammalian Circadian Clock. *Mol Cell*, 67, 770-782.e6. [🔗](#)
- Ye R, Selby CP, Chiou YY, Ozkan-Dagliyan I, Gaddameedhi S & Sancar A (2014). Dual modes of CLOCK:BMAL1 inhibition mediated by Cryptochrome and Period proteins in the mammalian circadian clock. *Genes Dev*, 28, 1989-98. [🔗](#)
- Cao X, Yang Y, Selby CP, Liu Z & Sancar A (2021). Molecular mechanism of the repressive phase of the mammalian circadian clock. *Proc Natl Acad Sci U S A*, 118. [🔗](#)

Langmesser S, Tallone T, Bordon A, Rusconi S & Albrecht U (2008). Interaction of circadian clock proteins PER2 and CRY with BMAL1 and CLOCK. BMC Mol Biol, 9, 41. [🔗](#)

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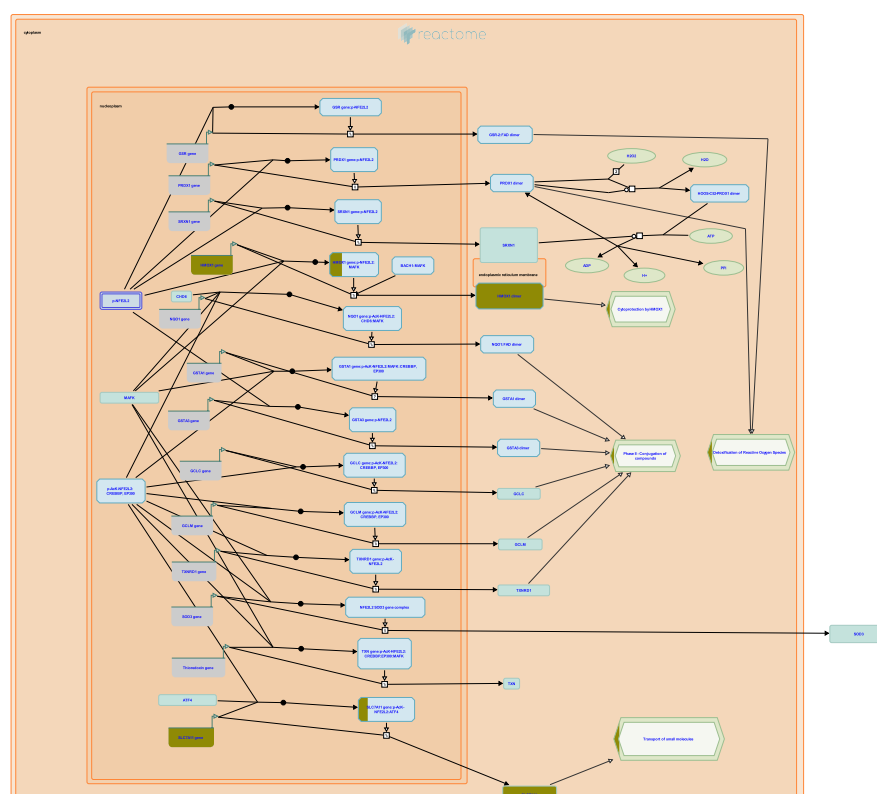
Date	Action	Author
2024-12-09	Edited	May B
2024-12-09	Authored	May B
2024-12-12	Created	May B
2025-03-04	Reviewed	Sancar A, Albrecht U

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
PER1	O15534

Input	Ensembl Id
PER1	ENSG00000179094

5. NFE2L2 regulating anti-oxidant/detoxification enzymes ([R-HSA-9818027](#))



Cellular compartments: nucleoplasm.

Subpathway representing cytoprotective genes regulated by NFE2L2 (NRF2). NFE2L2 is well-studied for its role in oxidative stress where it gets activated by ROS and then induces a plethora of gene expression regulation the oxidative damage. It induces genes/enzymes that regulate the phase 2 detoxification system (eg. GSTs and Glutathione system), ROS scavenging (SODs, PRDX1) and cytoprotection (HO1) by regulating inflammation and tissue damage (Tonelli et al, 2018; Shaw et al, 2020)

References

Edit history

Date	Action	Author
2022-10-10	Edited	Tiwari K
2022-10-10	Authored	Tiwari K
2022-10-10	Created	Tiwari K
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 4 Reactome entities

Input	UniProt Id	Input	UniProt Id
HMOX1	P09601	SLC7A11	Q9UPY5

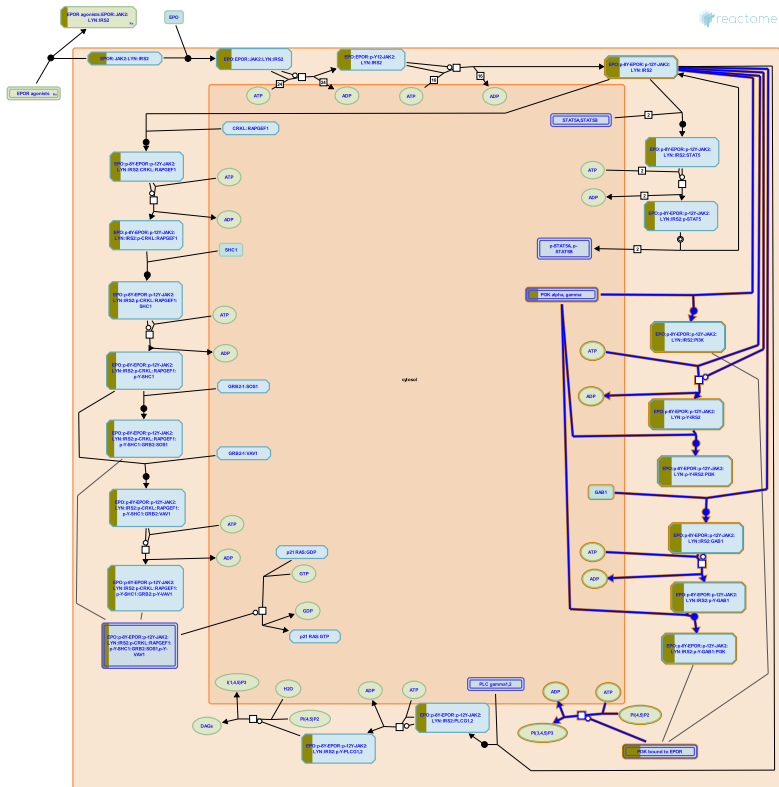
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HMOX1	ENSG000000100292	SLC7A11	ENSG000000151012

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with
CASP3	P42574	P10599
TXNIP	Q9H3M7	P10599

Input	UniProt Id	Interacts with
DDIT4L	Q96D03	P10599

6. Erythropoietin activates Phosphoinositide-3-kinase (PI3K) (R-HSA-9027276)



PI3K can bind the activated EPO receptor (EPOR) by three different mechanisms: direct binding to phospho-Y479 of the EPOR, indirect binding via phosphorylated IRS2 bound to the EPOR, and indirect binding via phosphorylated GAB1 bound to the EPOR (Bouscary et al. 2003, Schmidt et al. 2004, reviewed in Kuhrt and Wojchowski 2015). PI3K phosphorylates phosphatidylinositol 4,5-bisphosphate to yield phosphatidylinositol 3,4,5-trisphosphate which recruits AKT1 to the membrane.

References

Bouscary D, Pene F, Claessens YE, Muller O, Chrétien S, Fontenay-Roupie M, ... Lacombe C (2003). Critical role for PI 3-kinase in the control of erythropoietin-induced erythroid progenitor proliferation. Blood, 101, 3436-43.

Schmidt EK, Fichelson S & Feller SM (2004). PI3 kinase is important for Ras, MEK and Erk activation of Epo-stimulated human erythroid progenitors. BMC Biol., 2, 7.

Kuhrt D & Wojchowski DM (2015). Emerging EPO and EPO receptor regulators and signal transducers. Blood, 125, 3536-41.

Edit history

Date	Action	Author
2017-10-29	Edited	May B
2017-10-29	Authored	May B
2017-10-29	Created	May B
2018-08-14	Reviewed	McGraw KL

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
IRS2	Q9Y4H2

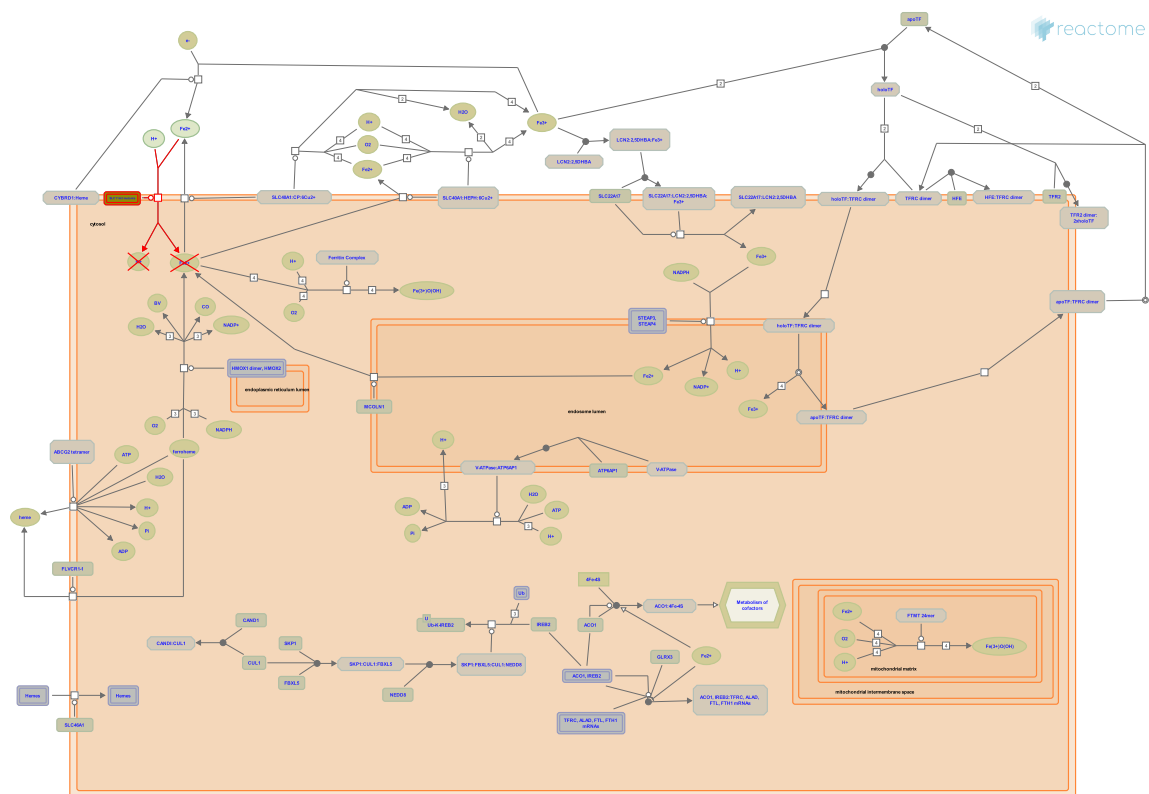
Input	UniProt Id
PIK3R1	P27986

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with
PIK3R1	P27986	Q13480

Input	UniProt Id	Interacts with
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7. Defective SLC11A2 causes hypochromic microcytic anemia, with iron overload 1 (AHMIO1) (R-HSA-5619048)



Diseases: hypochromic microcytic anemia.

The primary site for absorption of dietary iron is the duodenum. Ferrous iron (Fe^{2+}) is taken up from the gut lumen across the apical membranes of enterocytes and released into the portal vein circulation across basolateral membranes. The human gene SLC11A2 encodes the divalent cation transporter DCT1 (NRAMP2, Natural resistance-associated macrophage protein 2). DCT1 resides on the apical membrane of enterocytes and mediates the uptake of many metal ions, particularly ferrous iron, into these cells. Defects in SLC11A2 can cause hypochromic microcytic anemia, with iron overload 1 (AHMIO1; MIM:206100), a blood disorder characterised by high serum iron, large hepatic iron deposition, abnormal haemoglobin content in erythrocytes which are reduced in size and absence of sideroblasts and stainable bone marrow iron store (Shawki et al. 2012, Iolascon & De Falco 2009).

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Iolascon A & De Falco L (2009). Mutations in the gene encoding DMT1: clinical presentation and treatment. Semin. Hematol., 46, 358-70. [🔗](#)

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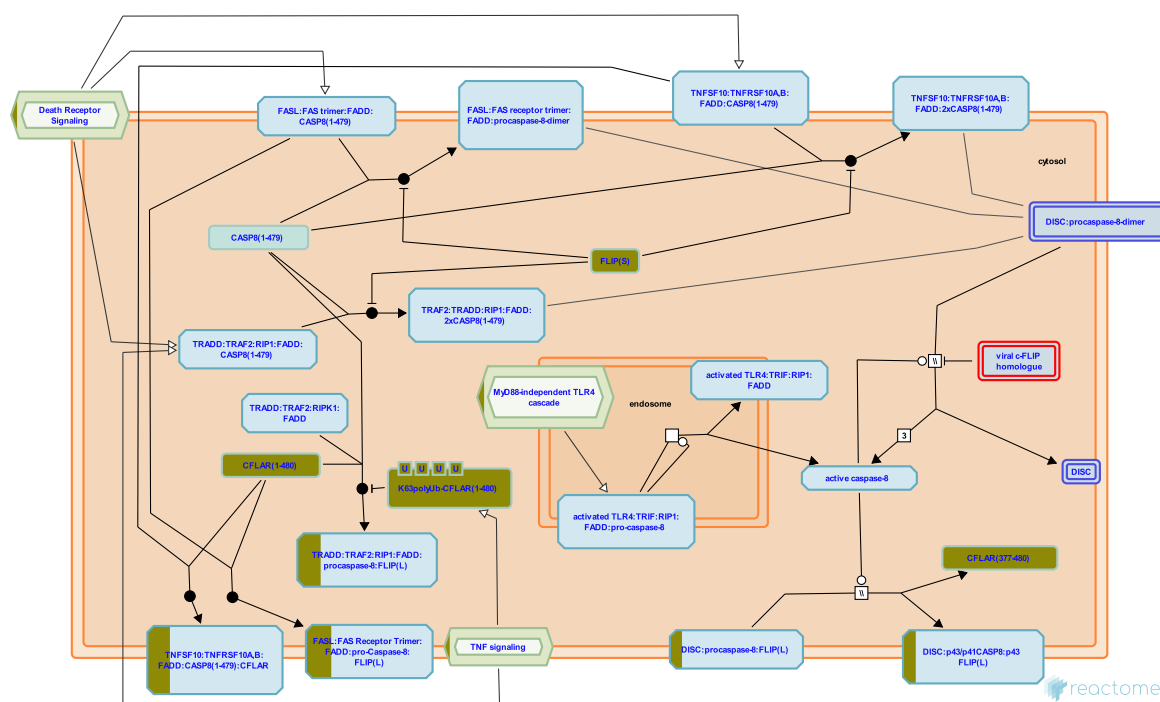
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2014-08-22	Edited	Jassal B

Date	Action	Author
2014-08-22	Authored	Jassal B
2014-08-22	Created	Jassal B
2015-08-04	Reviewed	Broer S
2023-10-12	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SLC11A2	P49281

8. Caspase activation via Death Receptors in the presence of ligand (R-HSA-140534)



Cellular compartments: cytosol.

Caspase-8 is synthesized as zymogen (procaspase-8) and is formed from procaspase-8 as a cleavage product. However, the cleavage itself appears not to be sufficient for the formation of an active caspase-8. Only the coordinated dimerization and cleavage of the zymogen produce efficient activation in vitro and apoptosis in cellular systems [Boatright KM and Salvesen GS 2003; Keller N et al 2010; Oberst A et al 2010].

The caspase-8 zymogens are present in the cells as inactive monomers, which are recruited to the death-inducing signaling complex (DISC) by homophilic interactions with the DED domain of FADD. The monomeric zymogens undergo dimerization and the subsequent conformational changes at the receptor complex, which results in the formation of catalytically active form of procaspase-8.[Boatright KM et al 2003; Donepudi M et al 2003; Keller N et al 2010; Oberst A et al 2010].

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Edit history

Date	Action	Author
2004-08-19	Created	Gopinathrao G
2012-11-19	Reviewed	Gillespie ME
2013-05-18	Edited	Shamovsky V
2013-05-22	Reviewed	Pop C, Salvesen GS
2015-01-12	Revised	Shamovsky V

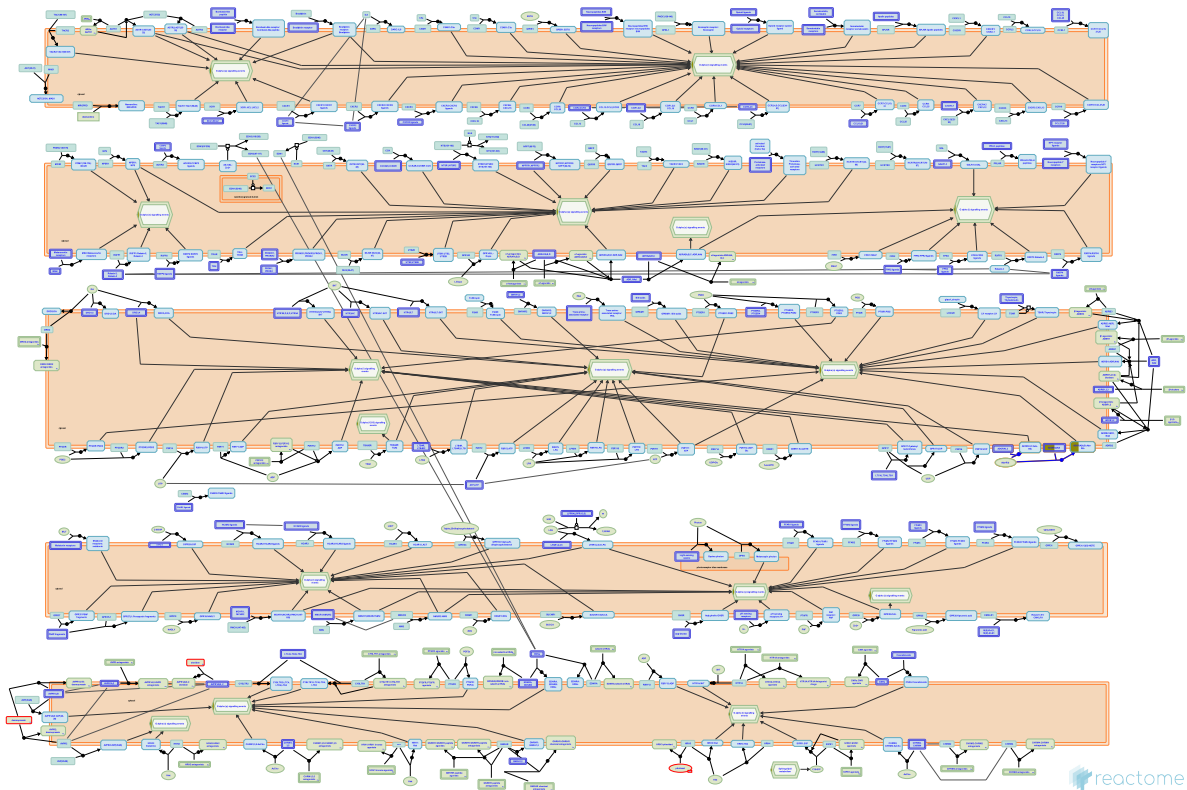
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CFLAR	O15519-1, O15519-2

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CFLAR	O15519, O15519-1	Q14790			

9. Adenosine P1 receptors (R-HSA-417973)



The adenosine receptors (P1 receptors) are a class of purinergic receptors, G-protein coupled receptors with adenosine as their endogenous ligand. In humans, there are four adenosine receptors. Each is encoded by a separate gene and the four receptors have distinct, though overlapping, functions. For instance, both A1 and A2A receptors play roles in the heart, regulating myocardial oxygen consumption and coronary blood flow. They also have important roles in the brain, regulating the release of other neurotransmitters such as dopamine and glutamate. The A2B and A3 receptors are located peripherally and are involved in processes such as inflammation and immune responses. Fredholm BB et al, 2001).

References

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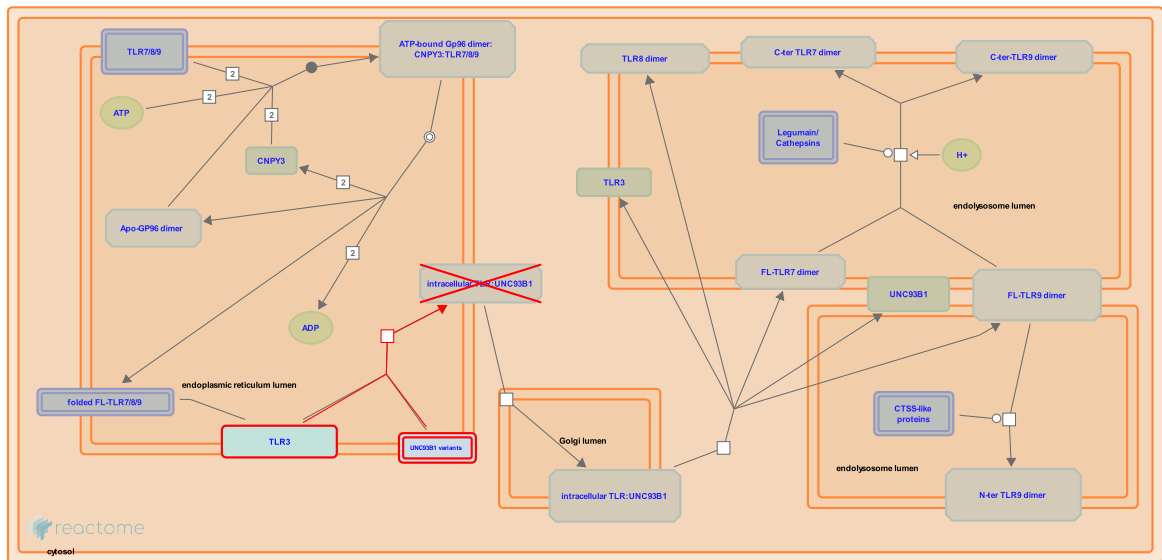
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Date	Action	Author
2009-04-14	Edited	Jassal B
2009-04-14	Authored	Jassal B
2009-04-14	Created	Jassal B
2009-05-29	Reviewed	D'Eustachio P
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
ADORA2B	P29275

10. UNC93B1 deficiency - HSE ([R-HSA-5602415](#))



Diseases: primary immunodeficiency disease.

UNC93B1 is an endoplasmic reticulum protein with 12 membrane-spanning domains. Signaling cascades of nucleotide-sensing endosomal toll like receptors (TLR3 and TLR7-9) depends on functional UNC93B1, which is thought to deliver these TLRs from the ER to the endosome where they recognize specific pathogenic patterns and initiate host immune responses.

UNC93B deficiency has been implicated in the increased susceptibility to herpes simplex virus type 1 (HSV1) encephalitis (HSE), a rare complication during HSV-1 infection of the central nervous system (CNS) (Casrouge A et al. 2006). Patients-derived UNC93B1-deficient fibroblasts showed an impaired production of IFN-beta and -gamma following stimulation with TLR3 agonist poly(I:C) (Casrouge A et al. 2006). These cells were also more susceptible to HSV1 infection, showing rapid viral replication together with high mortality rates. Furthermore, pluripotent stem cells (iPSC) derived from HSE patient dermal fibroblasts were differentiated into populations of neural stem cells (NSC), neurons, astrocytes and oligodendrocytes (Lafaille FG et al. 2012). The impaired induction of IFN beta and gamma was observed in all tested CNS cells upon stimulation with poly(I:C). However, HSV1 infection selectively affected type I and III IFN production in UNC93B1-deficient neurons and oligodendrocytes (Lafaille FG et al. 2012). Thus, impaired TLR3-mediated UNC93B-dependent type I and III IFN production in response to HSV1 infection in CNS, in neurons and oligodendrocytes in particular, may underline the pathogenesis of HSE in patients with UNC93B1 deficiency (Casrouge A et al. 2006; Lafaille FG et al. 2012).

Defective UNC93B1 also impairs the TLR7, TLR8 and TLR9 signaling pathways. Peripheral blood mononuclear cells (PBMCs) from UNC93B-deficient patients did not respond to the stimulation of TLR7, TLR8, or TLR9, in terms of the production of type I and III interferons, and other cytokines tested (Casrouge A et al. 2006). Moreover, no inducible CD62L shedding on granulocytes was detected after stimulation of whole blood cells derived from UNC93B-deficient patients with R-848 (agonist of TLR7 and TLR8) (von Bernuth H. et al. 2008). However, no clinical condition has been so far associated with impaired TLR7, TLR8, TLR9 due to UNC93B1 deficiency so this defect is not annotated here.

References

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Lafaille FG, Pessach IM, Zhang SY, Ciancanelli MJ, Herman M, Abhyankar A, ... Notarangelo LD (2012). Impaired intrinsic immunity to HSV-1 in human iPSC-derived TLR3-deficient CNS cells. *Nature*, 491, 769-73. [↗](#)

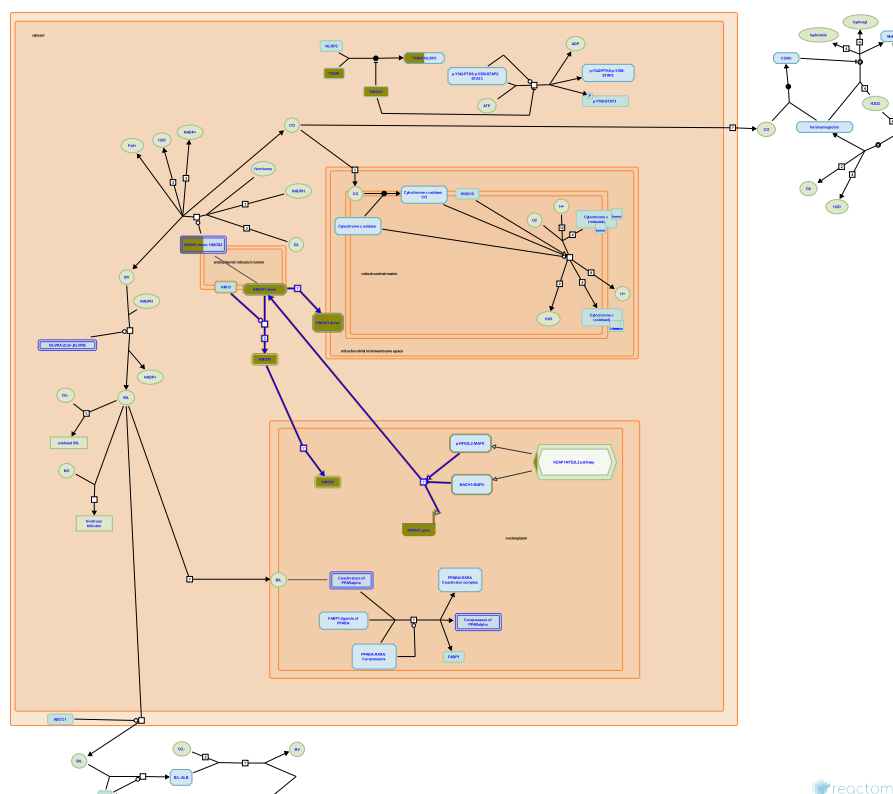
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2014-05-21	Authored	Shamovsky V
2014-06-24	Created	Shamovsky V
2014-09-06	Reviewed	D'Eustachio P
2015-02-10	Edited	Shamovsky V
2015-02-15	Reviewed	McDonald DR
2023-10-12	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PIK3R1	P27986	O15455			

11. Regulation of HMOX1 expression and activity (R-HSA-9707587)



Heme oxygenase 1 (HMOX1) is regulated at the level of gene transcription, mRNA translation, localization and degradation. Its gene is often activated under a wide range of stressful conditions. The transcriptional control of HMOX1 is determined by inducible regulatory elements localized in the 5' region of the promoter, so called antioxidant response elements (ARE)(Raghunath et al, 2018).

AREs on the HMOX1 gene are ultimately controlled by the enhancing NFE2L2:MAFK dimer and the repressing BACH1:MAFK dimer, both of which are influenced by a multitude of processes. Less specific enhancement occurs via AP-1 (FOS:JUN) dimers (Funes et al, 2020).

HMOX1 activity depends on dimerization in the ER membrane. Its membrane localization is abandoned by cleavage of the membrane domain by HM13. The resulting soluble enzyme is found in the cytosol, mitochondria, and the nucleus (Schaefer et al, 2017).

References

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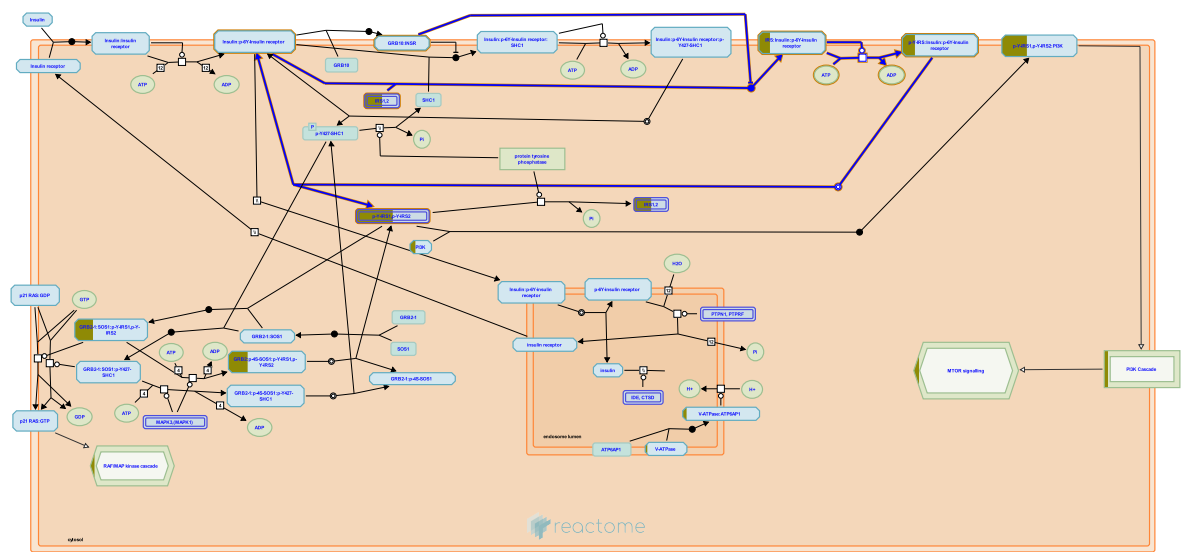
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2020-11-12	Edited	Stephan R
2020-11-12	Authored	Stephan R
2020-11-19	Created	Stephan R
2021-01-23	Reviewed	Somers J
2022-02-24	Revised	Rothfels K
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
HMOX1	P09601

Input	Ensembl Id
HMOX1	ENSG00000100292

12. IRS activation (R-HSA-74713)



IRS is one of the mediators of insulin signalling events. It is activated by phosphorylation and triggers a cascade of events involving PI3K, SOS, RAF and the MAP kinases. The proteins mentioned under IRS are examples of IRS family members acting as indicated. More family members are to be confirmed and added in the future.

Using receptor mutagenesis studies it is known that IRS1 via its PTB domain binds to the insulin receptor at the juxtamembrane region at tyrosine 972. The interaction is further stabilized by the PH domain of IRS1 which interacts with the phospholipids of the plasma membrane. This allows the receptor to phosphorylate IRS1 on up to 13 of its tyrosine residues. Once phosphorylated the IRS1 falls away from the receptor. Now in a tyrosine phosphorylated and hence activated state other proteins can interact with the IRS proteins.

References

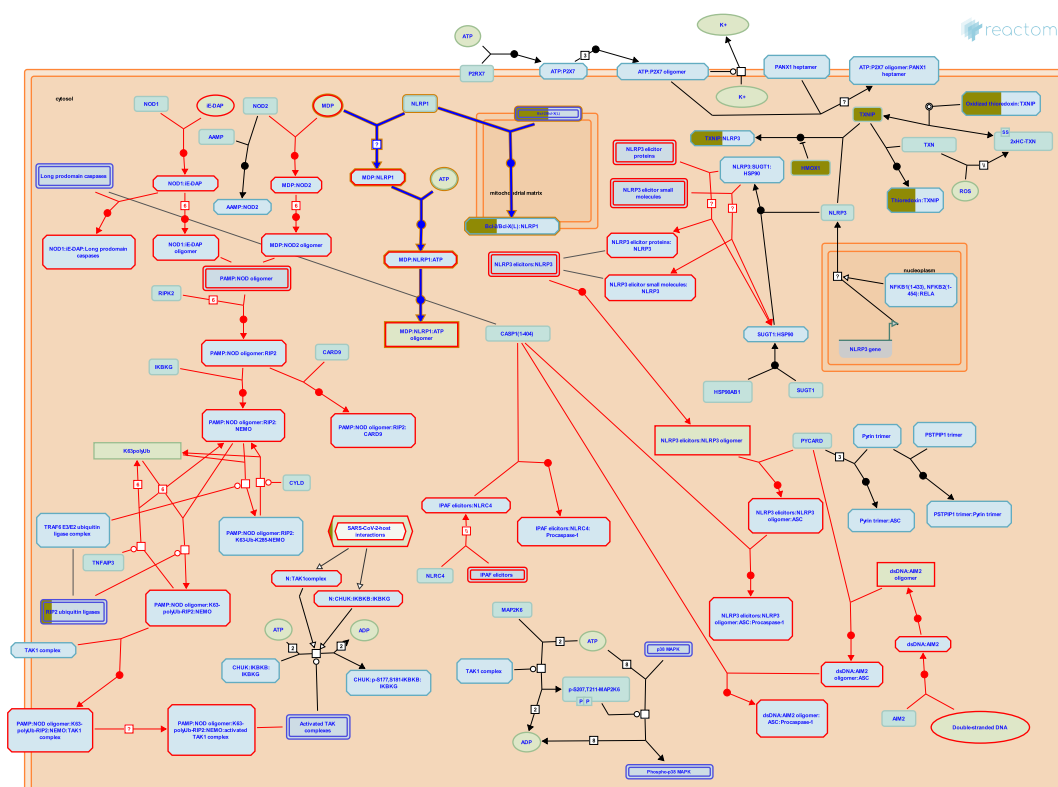
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Date	Action	Author
2003-07-28	Authored	
2003-07-31	Created	Bevan AP

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
IRS2	Q9Y4H2

13. The NLRP1 inflammasome (R-HSA-844455)



Cellular compartments: cytosol.

NLRP1 is activated by MDP (Faustin et al. 2007). The NLRP1 inflammasome was the first to be characterized. It was described as a complex containing NALP1, ASC, caspase-1 and caspase-5 (Martinon et al. 2002). Unlike NLRP3, NLRP1 has a C-terminal extension containing a CARD domain, which has been reported to interact directly with procaspase-1, obviating a requirement for ASC (Faustin et al. 2007), though ASC was found to augment the interaction. Mouse NLRP1 has no PYD domain and would therefore not be expected to interact directly with procaspase-1. Like the NLRP3 inflammasome, K⁺ efflux appears to be essential for caspase-1 activation (Wickliffe et al. 2008). Ribonucleoside triphosphates (NTPs) are required for NALP1-mediated caspase-1 activation with ATP being the most efficient, Mg²⁺ was also required (Faustin et al. 2007). The human NLRP1 gene has 3 paralogues in mouse that are highly polymorphic. Differences between mouse strains underlie susceptibility to anthrax lethal toxin (Boyden & Dietrich 2006).

References

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Edit history

Date	Action	Author
2010-04-22	Authored	Jupe S
2010-05-28	Created	Jupe S

Date	Action	Author
2011-04-28	Edited	Jupe S
2011-04-28	Reviewed	Kufer TA
2011-06-06	Reviewed	Rittinger K, Wong E
2025-11-15	Modified	Weiser JD

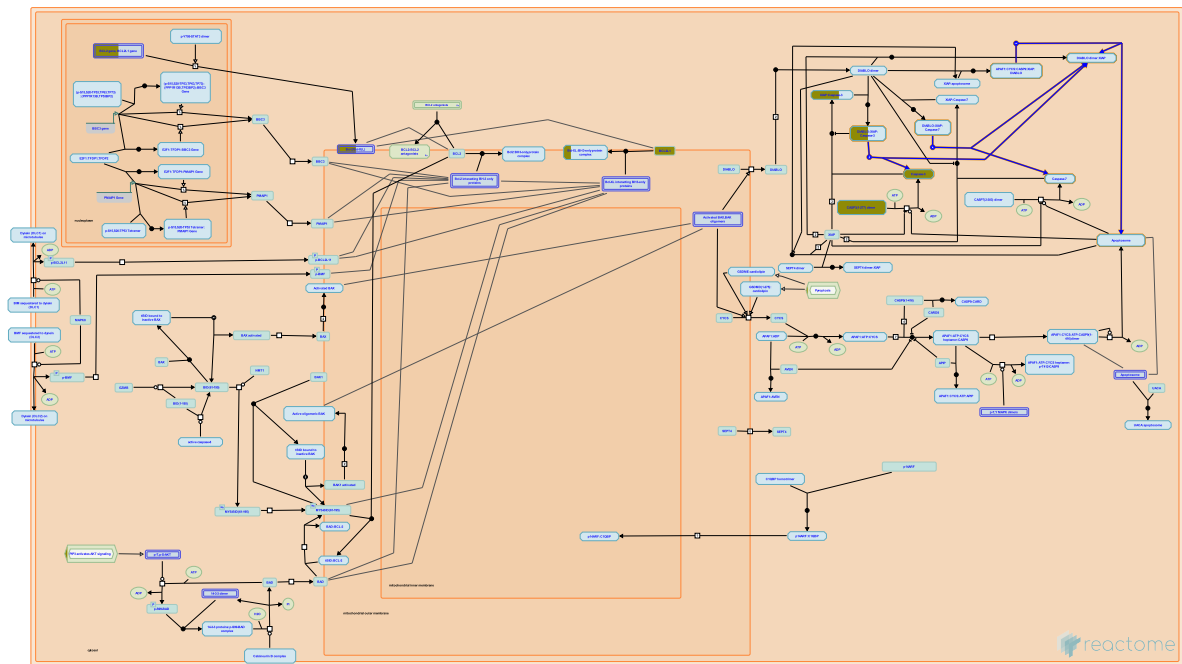
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BCL2L1	Q07817

Interactors found in this pathway (1)

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BCL2L1	Q07817	Q9C000			

14. SMAC(DIABLO)-mediated dissociation of IAP:caspase complexes (R-HSA-111464)



Cellular compartments: cytosol.

Second mitochondria derived activator of caspase/direct inhibitor of apoptosis binding protein with low pI (SMAC, also known as DIABLO) regulates XIAP function and potentiates caspase-3, -7 and -9 activity by disrupting the interaction of caspases with XIAP. Residues 56-59 of SMAC (DIABLO) are homologous to the amino-terminal motif that is used by caspase-9 (CASP9) to bind to the BIR3 domain of XIAP. SMAC (DIABLO) competes with CASP9 for binding to BIR3 domain of XIAP promoting the release of XIAP from the CASP9:apoptosome complex (Srinivasula SM et al. 2001; Salvesen et al. 2002). The binding of SMAC to the BIR2 and BIR3 regions of XIAP creates a steric hindrance that is essential for preventing binding of XIAP linker region with effector caspases CASP3 and CASP7 thus achieving neutralization of XIAP inhibition. The strong affinity for XIAP allows SMAC (DIABLO) to displace caspase-3, -7 from the XIAP:caspase complexes (Wu G et al. 2000; Chai J et al. 2001; Huang Y et al. 2003; Abhari BA & Davoodi J 2008).

References

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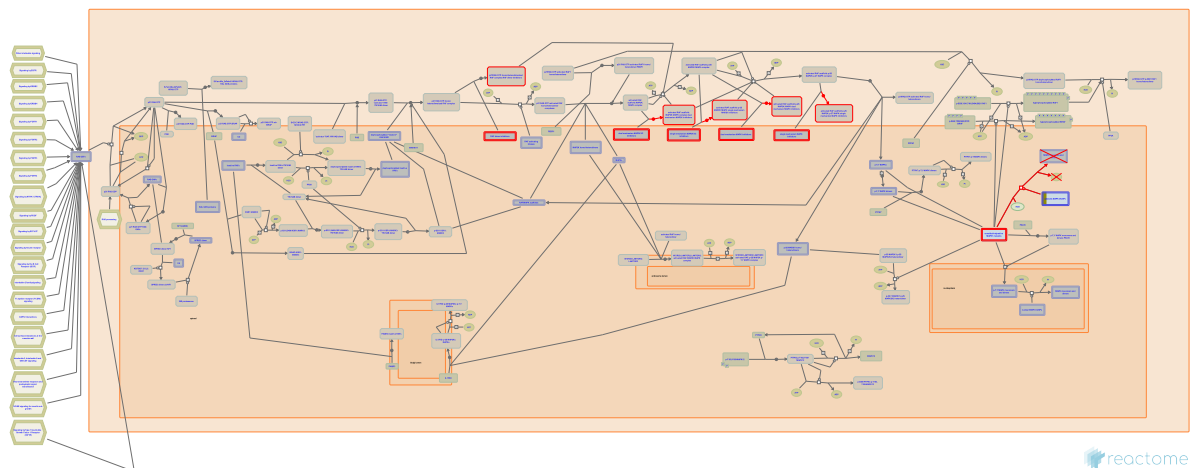
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Date	Action	Author
2004-02-17	Created	Alnemri E
2018-03-02	Reviewed	Matthews L
2018-11-02	Edited	Shamovsky V

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
CASP3	P42574

15. Signaling by MAPK mutants (R-HSA-9652817)



Diseases: cancer.

Mutations in ERK proteins (MAPK1 and MAPK2) are rare, with mutations reported in ~8% of cervical cancers and 1.5% of head and neck squamous cell carcinomas (Ojesina et al, 2014; Cancer Genome Atlas, 2015; reviewed in Najafi et al, 2019). MAPK proteins with activating mutations are often still dependent on upstream phosphorylation by MAP2K proteins, but support sustained downstream signaling by virtue of being resistant to inactivating dephosphorylations (reviewed in Samatar and Poulikakos, 2014; Liu et al, 2018).

References

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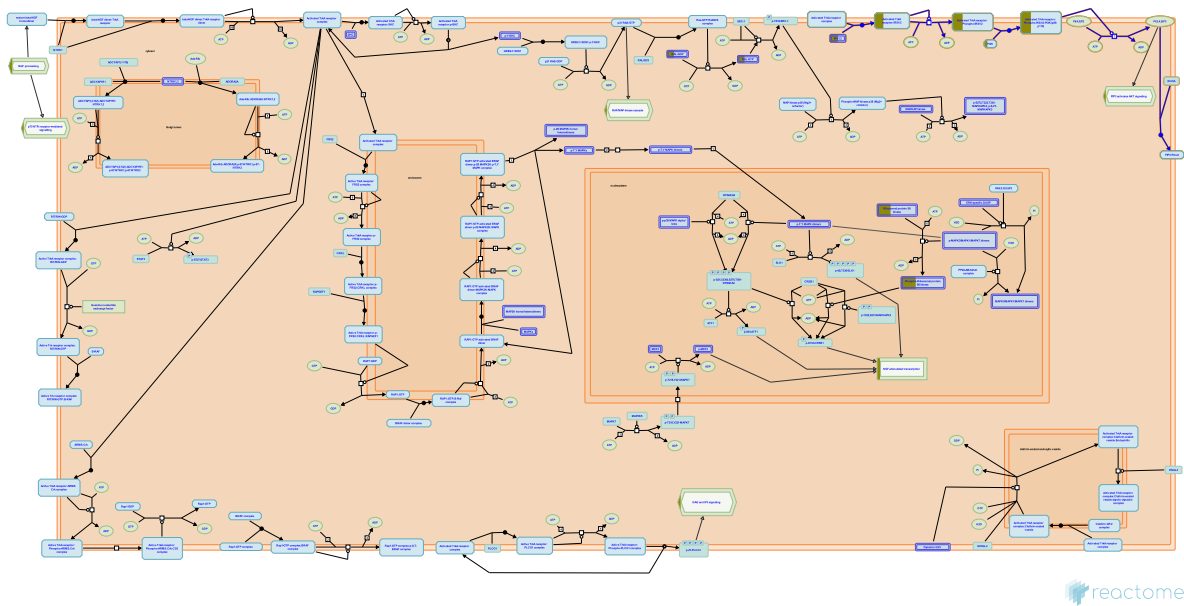
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2019-07-03	Created	Rothfels K
2019-10-25	Authored	Rothfels K
2020-05-04	Reviewed	Gavathiotis E
2020-05-26	Edited	Rothfels K

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
DUSP10	Q9Y6W6

16. PI3K/AKT activation (R-HSA-198203)



PI3K/AKT signalling is a major regulator of neuron survival. It blocks cell death by both impinging on the cytoplasmic cell death machinery and by regulating the expression of genes involved in cell death and survival. In addition, it may also use metabolic pathways to regulate cell survival. The PI3K/AKT pathway also affects axon diameter and branching (Marcus et al, 2002) and regulates small G proteins like RhoA (Vanhaesebroeck, B and Waterman, MD, 1999), which control the behaviour of the F-actin cytoskeleton. Moreover, through its connection with the TOR pathway, it promotes translation of a subset of mRNAs.

References

Markus A, Zhong J & Snider WD (2002). Raf and akt mediate distinct aspects of sensory axon growth . Neuron, 35, 65-76. [↗](#)

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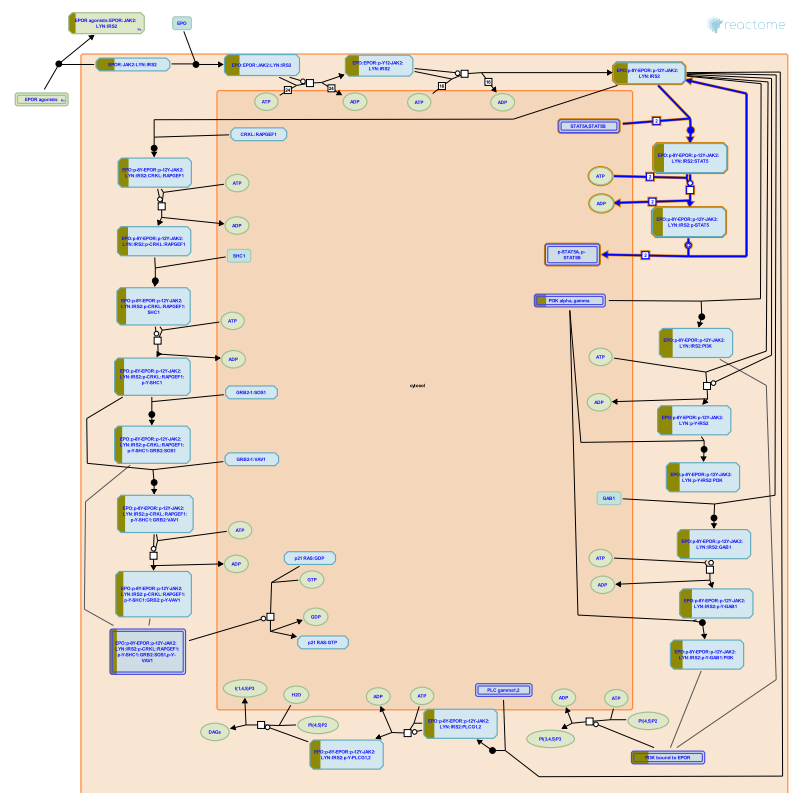
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2006-10-10	Authored	Annibali D, Nasi S
2007-05-21	Created	Jassal B
2007-11-08	Reviewed	Greene LA

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
IRS2	Q9Y4H2	PIK3R1	P27986

17. Erythropoietin activates STAT5 (R-HSA-9027283)



STAT5 (STAT5A or STAT5B) directly binds the phosphorylated cytoplasmic domain of EPOR, where it is phosphorylated by JAK2 and LYN (Oda et al. 1998, inferred from mouse homologs, reviewed in Kuhrt and Wojchowski 2015). Phosphorylated STAT5 then dissociates from EPOR, dimerizes, and transits to the nucleus where it activates gene expression.

References

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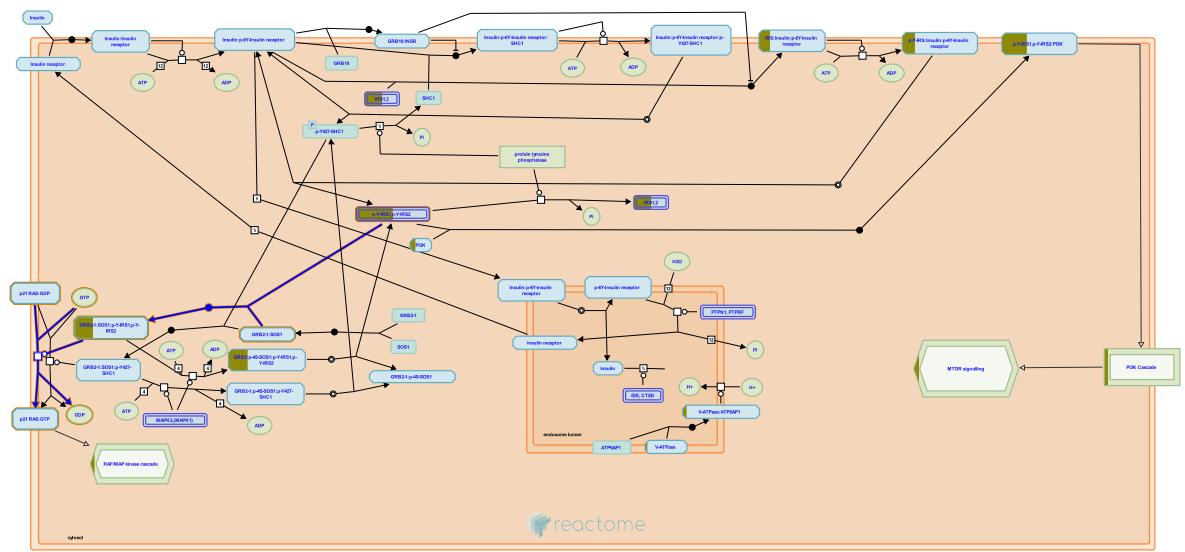
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2017-10-29	Edited	May B
2017-10-29	Authored	May B
2017-10-29	Created	May B
2018-08-14	Reviewed	McGraw KL

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
IRS2	Q9Y4H2

18. SOS-mediated signalling (R-HSA-112412)



Cellular compartments: plasma membrane, cytosol.

SOS is recruited to the plasma membrane and mediates activation of Ras.

References

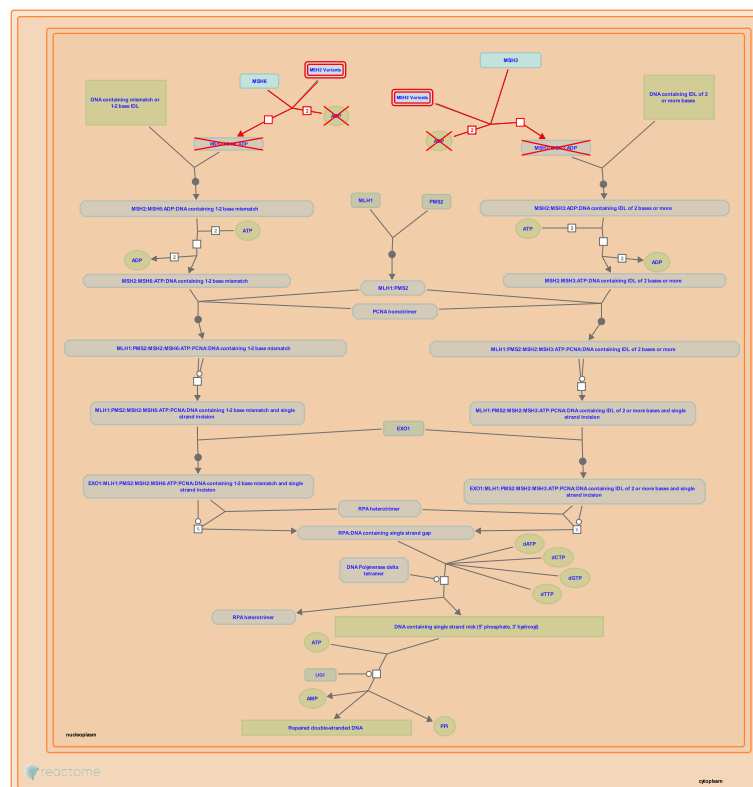
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Date	Action	Author
2004-04-29	Authored	Charalambous M
2004-04-29	Created	Charalambous M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
IRS2	Q9Y4H2

19. Defective Mismatch Repair Associated With MSH2 ([R-HSA-5632928](#))



Cellular compartments: nucleoplasm.

Diseases: cancer.

MSH2 is homologous to the *E. coli* MutS gene and is involved in DNA mismatch repair (MMR) (Fishel et al., 1994). Heterozygous mutations in the MSH2 gene result in hereditary nonpolyposis colorectal cancer-1. Variants of MSH2 are associated with hereditary nonpolyposis colorectal cancer. Alteration of MSH2 is also involved in Muir-Torre syndrome and mismatch repair cancer syndrome.

References

Mitchell RJ, Farrington SM, Dunlop MG & Campbell H (2002). Mismatch repair genes hMLH1 and hMSH2 and colorectal cancer: a HuGE review. *Am. J. Epidemiol.*, 156, 885-902. [🔗](#)

Fishel R, Lescoe MK, Rao MR, Copeland NG, Jenkins NA, Garber J, ... Kolodner R (1993). The human mutator gene homolog MSH2 and its association with hereditary nonpolyposis colon cancer. *Cell*, 75, 1027-38. [🔗](#)

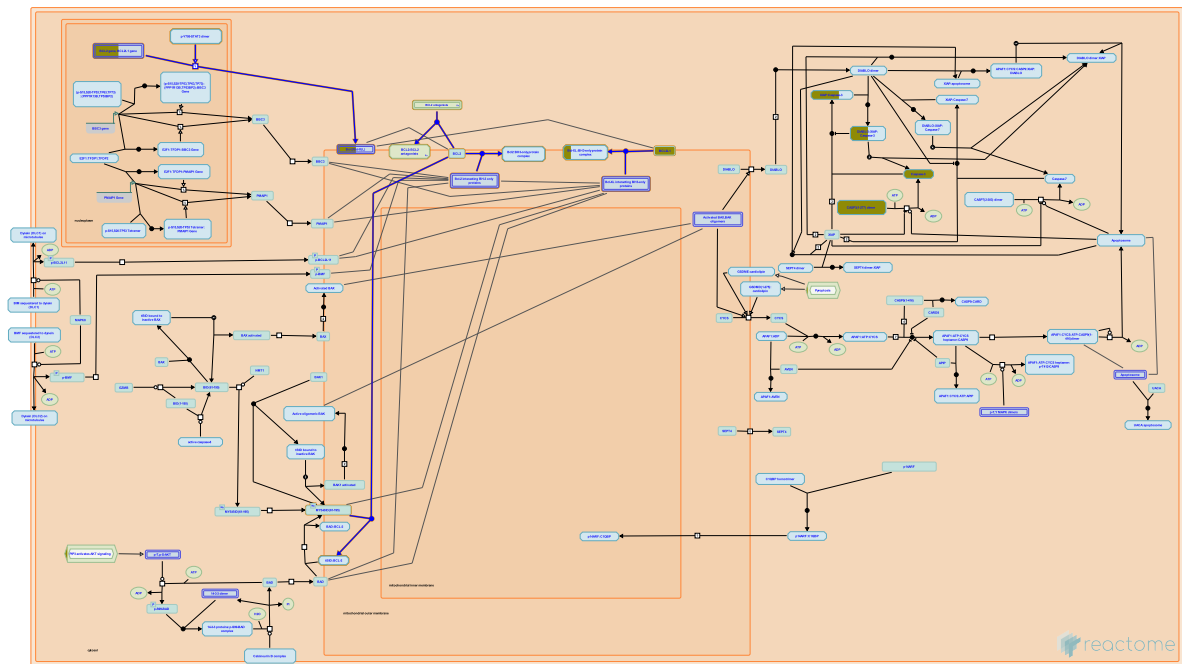
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Date	Action	Author
2011-11-10	Authored	Gillespie ME
2014-10-29	Created	Gillespie ME
2016-11-01	Reviewed	Arora S
2017-02-27	Edited	Gillespie ME
2023-03-08	Modified	Matthews L

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MYC	P01106	P52701			

20. BH3-only proteins associate with and inactivate anti-apoptotic BCL-2 members (R-HSA-111453)



Cellular compartments: mitochondrial outer membrane.

Bcl-2 interacts with tBid (Yi et al. 2003), BIM (Puthalakath et al. 1999), PUMA (Nakano and Vousden 2001), NOXA (Oda et al. 2000), BAD (Yang et al. 2005), BMF (Puthalakath et al. 2001), resulting in inactivation of BCL2. Binding of BCL2 to tBID inhibits BID-induced cytochrome C release and apoptosis (Yi et al. 2003). BH3 only proteins associate with and inactivate anti-apoptotic BCL-XL.

References

Edit history

Date	Action	Author
2004-08-09	Created	Tsujimoto Y, Hardwick JM

1 submitted entities found in this pathway, mapping to 2 Reactome entities

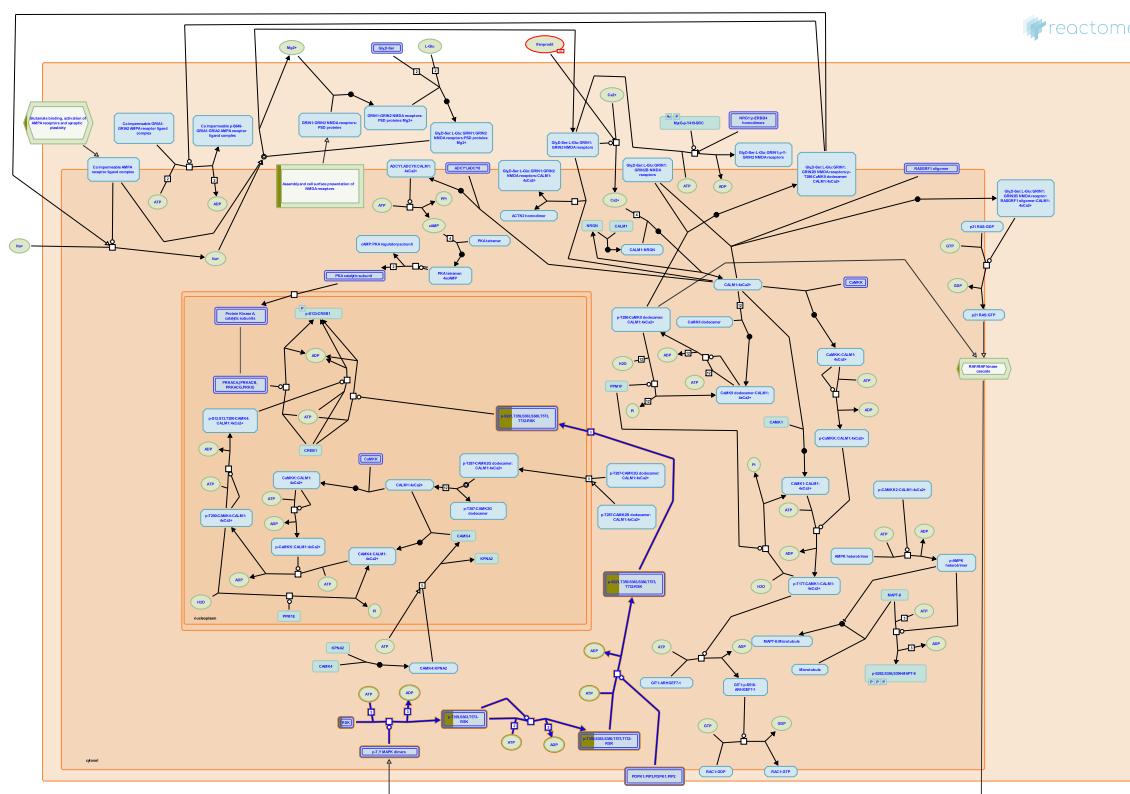
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BCL2L1	Q07817

Input	Ensembl Id
BCL2L1	ENSG00000171552

Interactors found in this pathway (2)

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BCL2L1	Q07817, Q07817-1	Q07817, P55957	DDIT4L	Q96D03	P55957

21. RSK activation (R-HSA-444257)



Cellular compartments: cytosol, nucleoplasm.

Ribosomal S6 kinase (RSK) has four isoforms in humans, RPS6KA1 (RSK1), RPS6KA2 (RSK3), RPS6KA3 (RSK2) and RPS6KA6 (RSK4), and each of the isoforms have six conserved phosphorylation sites (in RPS6KA1, these are serine residues S221, S363 and S380 and threonine residues T359, T573 and T732). Phosphorylation at four of these residues appears to be critically important for the catalytic activity of RSKs: S221, S363, S380 and T573 (in RPS6KA1).

Phosphorylation and activation of RSKs primarily occurs at the plasma membrane, but can occur in the cytoplasm and in the nucleus. ERKs (MAPK1 and MAPK3), activated downstream of RAS signaling, phosphorylate RSKs on threonine and serine residues T359, S363 and T573 (in RPS6KA1). Phosphorylation by ERKs enables autophosphorylation of RSKs on serine residue S380 and threonine residue T732 (in RPS6KA1). Phosphorylation of RSKs by PDK1 (PDK1) at serine residue S221 (in RPS6KA1) is necessary for the full activation of RSKs and phosphorylation of RSK substrates (reviewed by Anjum and Blenis 2012). RSK4 differs from other RSKs because it shows high level of constitutive phosphorylation and activity in the absence of growth factors, although it is still responsive to growth factors and ERK activity (Dummler et al. 2005).

RSKs, especially RSK2, are highly expressed in brain regions with high synaptic activity. RSK2 mutations are the underlying cause of Coffin-Lowry syndrome (CLS), which is characterized by cognitive impairment and skeletal anomalies (Zeniou et al. 2002).

References

Anjum R & Blenis J (2008). The RSK family of kinases: emerging roles in cellular signalling. *Nat Rev Mol Cell Biol*, 9, 747-58. [🔗](#)

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Zeniou M, Ding T, Trivier E & Hanauer A (2002). Expression analysis of RSK gene family members: the RSK2 gene, mutated in Coffin-Lowry syndrome, is prominently expressed in brain structures essential for cognitive function and learning. Hum Mol Genet, 11, 2929-40. [↗](#)

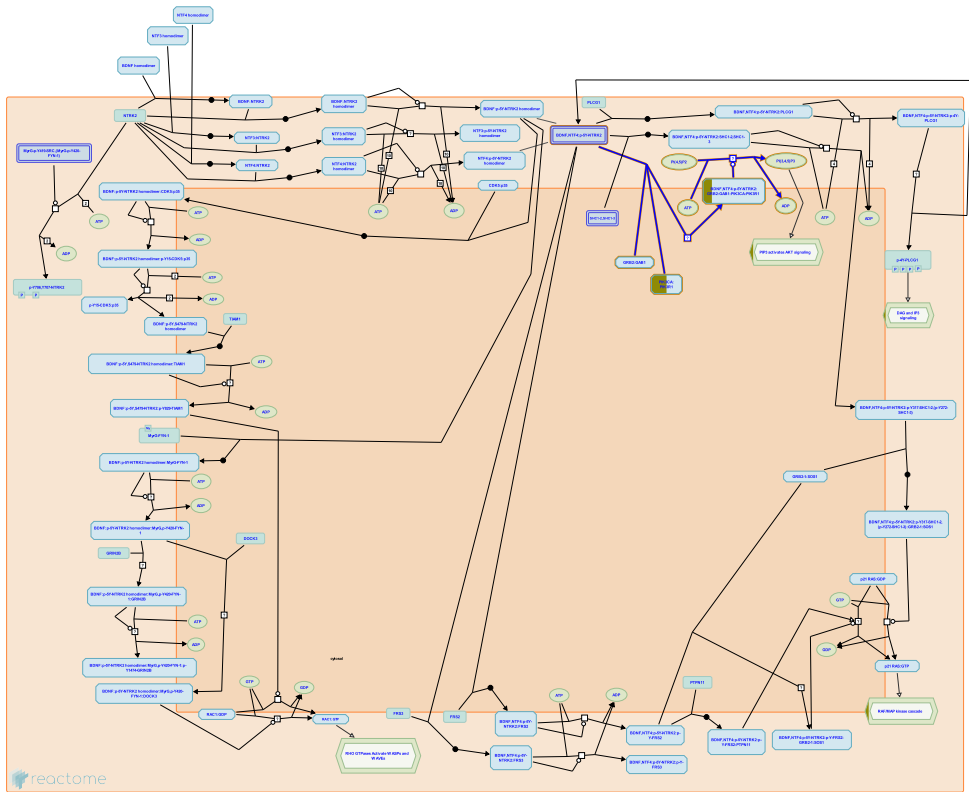
Edit history

Date	Action	Author
2009-10-23	Created	Mahajan SS
2009-10-29	Authored	Mahajan SS
2009-11-18	Reviewed	Tukey D
2009-11-19	Edited	Gillespie ME
2018-10-10	Revised	Orlic-Milacic M
2018-11-02	Reviewed	Hansen KB, Yi F
2018-11-07	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPS6KA3	P51812

22. Activated NTRK2 signals through PI3K (R-HSA-9028335)



Neurotrophin receptor NTRK2 (TRKB), activated by BDNF or NTF4, activates PI3K, resulting in formation of the PIP3 secondary messenger. PIP3 activates AKT signaling, and AKT signaling activates mTOR signaling (Yuen and Mobley 1999, Cao et al. 2013).

References

Yuen EC & Mobley WC (1999). Early BDNF, NT-3, and NT-4 signaling events. *Exp. Neurol.*, 159, 297-308. [🔗](#)

Cao C, Rioult-Pedotti MS, Migani P, Yu CJ, Tiwari R, Parang K, ... Marshall J (2013). Impairment of TrkB-PSD-95 signaling in Angelman syndrome. *PLoS Biol.*, 11, e1001478. [🔗](#)

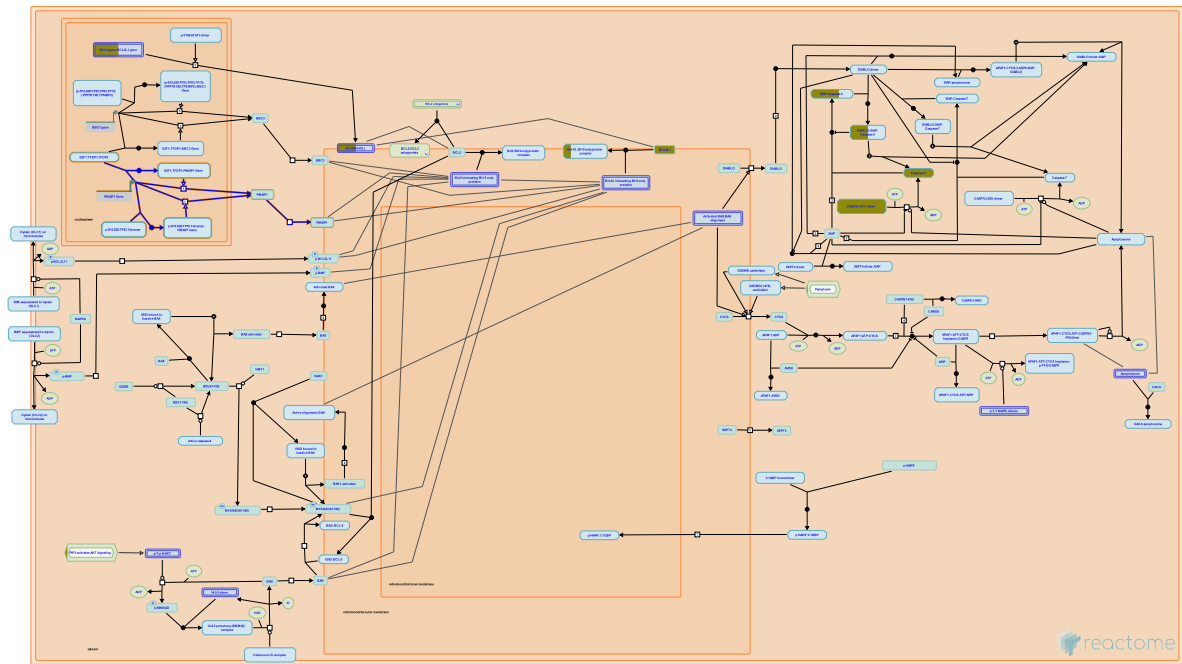
Edit history

Date	Action	Author
2017-11-07	Created	Orlic-Milacic M
2018-01-05	Authored	Orlic-Milacic M
2018-02-13	Reviewed	Antila H, Castrén E
2018-02-20	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
PIK3R1	P27986

23. Activation of NOXA and translocation to mitochondria (R-HSA-111448)



Cellular compartments: cytosol.

NOXA is transactivated in a p53-dependent manner and by E2F1. Activated NOXA is translocated to mitochondria.

References

Li CQ, Robles AI, Hanigan CL, Hofseth LJ, Trudel LJ, Harris CC & Wogan GN (2004). Apoptotic signaling pathways induced by nitric oxide in human lymphoblastoid cells expressing wild-type or mutant p53. *Cancer Res*, 64, 3022-9. [🔗](#)

Hershko T & Ginsberg D (2004). Up-regulation of Bcl-2 homology 3 (BH3)-only proteins by E2F1 mediates apoptosis. *J Biol Chem*, 279, 8627-34. [🔗](#)

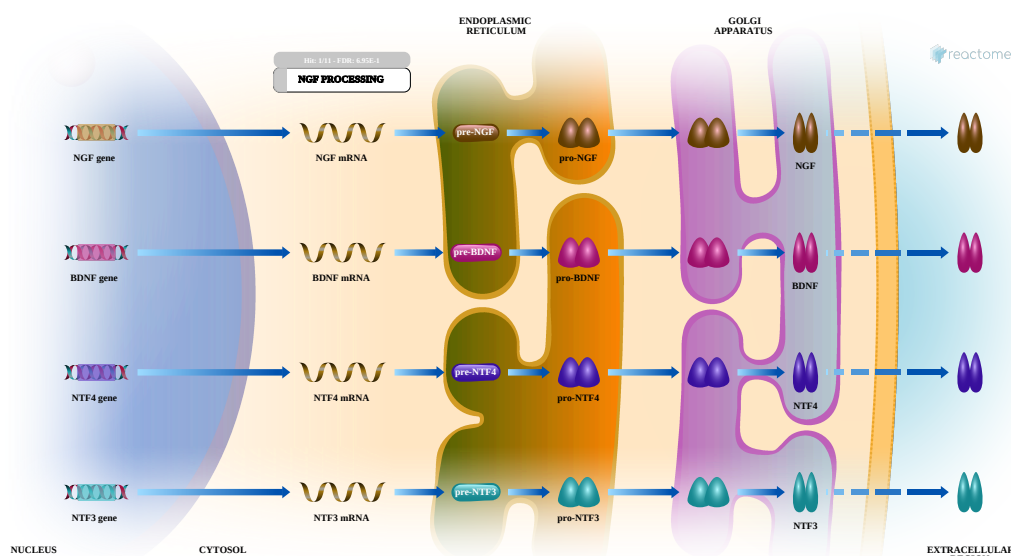
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Date	Action	Author
2004-08-09	Created	Tsujimoto Y, Hardwick JM

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BCL2L1	Q07817-1	Q13794			

24. Expression and Processing of Neurotrophins (R-HSA-9036866)



Neurotrophins function as ligands for receptor tyrosine kinases of the NTRK (TRK) family, as well as the death receptor NGFR (p75NTR). While all four neurotrophins, NGF, BDNF, NTF3 (NT-3) and NTF4 (NT-4, NT-5, NTF5) can bind to and activate NGFR, they show different specificity for NTRKs. NGF exclusively activates NTRK1 (TRKA). BDNF and NTF4 are high affinity ligands for NTRK2 (TRKB). NTF3 is a high affinity ligand for NTRK3 (TRKC) and a low affinity ligand for NTRK2. Neurotrophins play pivotal roles in survival, differentiation, and plasticity of neurons in the peripheral and central nervous system. They are produced, and secreted in minute amounts, by a variety of tissues. For review, please refer to Lessmann et al. 2003, Chao 2003, and Park and Poo 2013.

Human NGF, also known as the nerve growth factor, is encoded by a gene on chromosome 1, which produces a single transcript. Nascent NGF protein, pre-pro-NGF, is 241 amino acids long. As pre-pro-NGF enters the endoplasmic reticulum (ER), the signal peptide, consisting of eighteen amino acids at the N-terminus, is cleaved, producing pro-NGF. Two molecules of pro-NGF form homodimers in the ER. After transport of pro-NGF homodimers to the Golgi, 103 amino acids at the N-terminus of pro-NGF are cleaved, producing mature NGF homodimers. Both pro-NGF homodimers and mature NGF homodimers are secreted to the extracellular space. Mature NGF homodimers activate NTRK1 signaling, while NGFR signaling can be activated by both mature and pro-NGF homodimers. Secreted pro-NGF homodimers may be cleaved by extracellular matrix proteases to produce mature NGF homodimers. For review, please refer to Poo 2001, Lu et al. 2005, Skaper et al. 2012, Bradshaw et al. 2015.

Human BDNF, also known as brain-derived neurotrophic factor, is encoded by a gene on chromosome 11, which, through the use of 9 alternative promoters and alternative splicing, produces 17 protein-coding transcripts. Most BDNF transcripts result in the same pre-pro-BDNF protein of 247 amino acids, but alternative promoters and different 5' and 3'UTRs allow to fine-tune regulation of BDNF expression at different developmental stages and at different levels of neuronal activity. Similar to NGF, pre-pro-BDNF is processed by proteolytic cleavage in the ER to produce pro-BDNF homodimers. It is unclear whether proteolytic processing of pro-BDNF, to produce mature BDNF homodimers, occurs in the Golgi or in the secretory granules. Extracellular matrix proteases can also cleave secreted pro-BDNF to produce mature BDNF homodimers. Secreted mature BDNF homodimers can activate NTRK2 signaling, while secreted pro-BDNF homodimers can activate NGFR signaling. For review, please refer to Poo 2001, Lu et al. 2005, Skaper et al. 2012, Park and Poo 2013.

Human NTF4, also known as neurotrophin-4, is transcribed from a gene on chromosome 19. A single experimentally confirmed transcript produces a pre-pro-NTF4 protein of 210 amino acids. After proteolytic processing in the ER and Golgi, mature NTF4 homodimers are secreted and can activate NTRK2 signaling (Hibbert et al. 2003). For review, please refer to Poo 2001, Skaper et al. 2012.

Human NTF3, also known as neurotrophin-3, is transcribed from a gene on chromosome 12. Two NTF3 transcripts have been experimentally confirmed, but only the longer NTF3 splice variant of 270 amino acids has been studied. After proteolytic processing in the ER and Golgi, mature NTF3 homodimers are secreted and can activate NTRK3 signaling (Seidah et al. 1996, Farhadi et al. 2000). For review, please refer to Poo 2001, Skaper et al. 2012.

References

- Lessmann V, Gottmann K & Malsangio M (2003). Neurotrophin secretion: current facts and future prospects. *Prog Neurobiol*, 69, 341-74. [↗](#)
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- Park H & Poo MM (2013). Neurotrophin regulation of neural circuit development and function. *Nat. Rev. Neurosci.*, 14, 7-23. [↗](#)
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- Lu B, Pang PT & Woo NH (2005). The yin and yang of neurotrophin action. *Nat. Rev. Neurosci.*, 6, 603-14. [↗](#)

Edit history

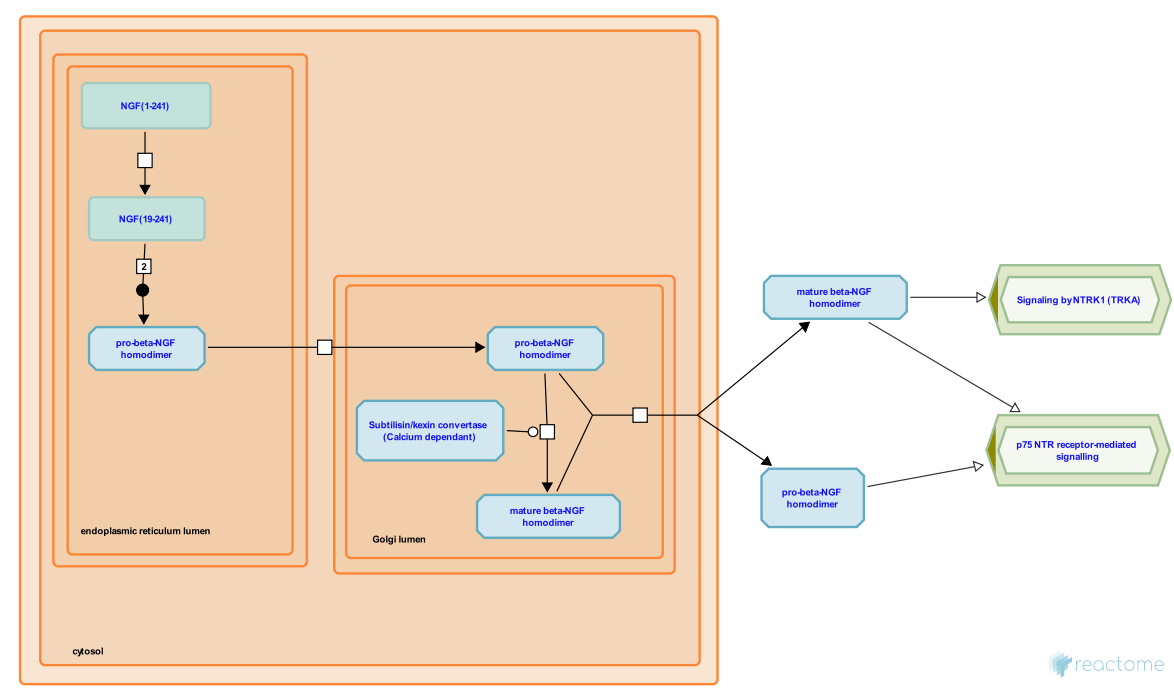
Date	Action	Author
2018-02-10	Edited	Orlic-Milacic M
2018-02-10	Authored	Orlic-Milacic M
2018-02-10	Created	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SORT1	Q99523	P01138			

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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25. NGF processing (R-HSA-167060)



All neurotrophins (NTs) are generated as pre-pro-neurotrophin precursors. The signal peptide is cleaved off as NT is associated with the endoplasmic reticulum (ER). The resulting pro-NT can form a homodimer spontaneously which then transits to the Golgi apparatus and then onto the trans-Golgi network (TGN). Resident protein convertases (PCs) can cleave off the pro-sequence and mature NT is targeted to constitutively released vesicles. The pro-NT form can also be released to the extracellular region.

References

Lessmann V, Gottmann K & Malsangio M (2003). Neurotrophin secretion: current facts and future prospects. Prog Neurobiol, 69, 341-74. [↗](#)

Edit history

Date	Action	Author
2006-02-07	Created	
2006-10-10	Edited	Jassal B
2006-10-10	Authored	Annibali D, Nasi S
2007-11-08	Reviewed	Greene LA
2025-11-15	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SORT1	Q99523	P01138			

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

49 of the submitted entities were found, mapping to 79 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ADORA2B	P29275	APOOL	Q6UXV4	ARL4C	P56559
ARMC8	Q8IUR7	ATP6V1A	P38606	BCL2L1	Q07817
BIRC3	Q13489	CASP3	P42574	CERS6	Q6ZMG9
CFLAR	O15519-1, O15519-2	COMMD7	Q86VX2	CPNE7	Q9UBL6
CRISPLD2	Q9H0B8	CSF1	P09603	DUSP10	Q9Y6W6
EHHADH	Q08426	EPHA4	P54764	FGD4	Q96M96
FOXP1	Q9H334-8	GAS1	P54826	GCNT1	Q02742
GEM	P55040	HMOX1	P09601	IFNGR1	P15260
IRS2	Q9Y4H2	LRRC8A	Q8IWT6	MBOAT2	Q6ZWT7
MGLL	Q99685	MIB1	Q86YT6	MTHFD2	P13995
MYC	P01106	NEK7	Q8TDX7	NIPAL2	Q9H841
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	PER1	O15534	PI4K2A	Q9BTU6
PIK3R1	P27986	RALA	P11233	RPS6KA3	P51812
SEMA3A	Q14563	SLC11A2	P49281	SLC7A11	Q9UPY5
SORT1	Q99523	SOX13	Q9UN79	TAF4B	Q92750
TGIF1	Q15583	TXNIP	Q9H3M7	UPP1	Q16831
ZEB1	P37275				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ARL4C	ENSG00000188042, ENST00000339728	ATP6V1A	ENSG00000114573	BCL2L1	ENSG00000171552
CSF1	ENSG00000184371	EPHA4	ENSG00000116106	GEM	ENSG00000164949
HMOX1	ENSG00000100292	MGLL	ENSG00000074416	MYC	ENSG00000136997
PER1	ENSG00000179094	PIK3R1	ENSG00000145675	RALA	ENSG00000006451
SLC7A11	ENSG00000151012	ZEB1	ENSG00000148516		

Interactors (55)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ADORA2B	P29275	P29274	AHDC1	Q5TGY3	Q13185
APBB2	Q92870-2	O15534	APOOL	Q6UXV4	P46059
ARL4C	P56559	P52306	ARMC8	Q8IUR7	Q6VN20, Q96S59
ATP6V1A	P38606	Q9UI12	BCL2L1	Q07817	Q9C000
BIRC3	Q13489	Q13546, Q9Y572	CASP3	P42574	P10599
CDK17	Q00537	P06493	CERS6	Q6ZMG9	P08173
CFLAR	O15519, O15519-1	Q14790	CPNE7	Q9UBL6-2	Q9NRD5
CSF1	P09603	P09603, P07333	DDIT4L	Q96D03	P10599
DUSP10	Q9Y6W6	P60709	EHHADH	Q08426	Q8IWP5
ENC1	O14682	Q13501	EPHA4	P54764	P05177
FGD4	Q96M96	P17252	FOXN2	P32314	Q9Y297
FOXP1	Q9H334	Q9BZS1	GCNT1	Q02742	Q07954

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GEM	P55040	Q9BUH8	HMOX1	P09601	P0DTC3
IFNGR1	P15260	P23458	IRS2	Q9Y4H2	Q96EB6
LRRC8A	Q8IWT6	O43914	MAP1A	P78559	Q9GZQ8
MGLL	Q99685	P07550	MIB1	Q86YT6	Q9UHD2
MTHFD2	P13995	P49137	MYC	EBI-1265559	P23771
NEK7	Q8TDX7	P00558	NR3C1	P06536	Q15788
NUDT6	P53370	P46199	PER1	J3QSH9, O15534	P56645, Q16526, O15055
PHLDB2	Q86SQ0	P21333	PI4K2A	Q9BTU6	Q96J02
PIK3R1	P27986-2, P27986	P48023, O15524	PLEKHA7	Q6IQ23-2	Q15669
RALA	P11233	P13569	RALGPS2	Q86X27	P63104
RPS6KA3	P51812	P08238	SLC11A2	P49281-3	P00403
SLC7A11	Q9UPY5	P16070	SNX30	Q8WV41	P48023
SORT1	Q99523	P01138	SOX13	Q9UN79	Q12933
TCEAL4	Q96EI5	Q93009	TGIF1	Q15583	Q9Y2T1
TXNIP	Q9H3M7	P10599	USP53	Q70EK8	P46109
ZEB1	P37275	Q92769			

7. Identifiers not found

These 7 identifiers were not found neither mapped to any entity in Reactome.

CAMK2N1

EPG5

FRMD4B

GK5

GSAP

HPS5

KCTD2